Violence Reduction, Firms, and Local Labor Markets: Evidence from a Ceasefire

Eleno Castro[1]

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Abstract

Can reductions in lethal violence expand economic activity even when criminal organizations remain active? I study El Salvador's 2012 gang truce using administrative data and a difference-in-differences design comparing areas with different pre-truce gang exposure. The number of registered firms increases by about 3 percent and formal employment by roughly 6 percent. Gains are concentrated among smaller firms and in non-tradable sectors. Fewer recorded exits suggest improved business survival. Firm-count gains largely reverse as violence resurges. Together, these findings highlight the economic benefits of lower violence and their vulnerability to renewed insecurity.

Keywords: Organized crime, homicides, labor markets, local demand, firms, economic growth

JEL codes: D74, K42, O17

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1 Introduction

How do firms and local labor markets respond when lethal violence falls sharply but criminal organizations remain active? A decline in killings may ease some constraints on economic activity even as other forms of criminal coercion persist. This question is particularly relevant in Latin America and the Caribbean, where homicide rates exceed the global average by more than threefold and organized criminal groups account for roughly half of all killings (United Nations Office on Drugs and Crime (UNODC), 2023).

This paper studies the economic effects of the sharp decline in homicides following the start of the Salvadoran gang truce in March 2012. Secretly negotiated between incarcerated gang leaders and government officials, the agreement cut homicide rates by an estimated 60% within weeks. Using monthly administrative records, I find larger increases in the number of registered firms and formal employment following the truce in areas with pre-truce gang exposure.

I combine monthly social security records on registered firms and employment with police homicide records. I measure pre-truce gang exposure using an indicator for whether a local administrative area, or polygon, recorded gang-related homicides before the truce. I then use a difference-in-differences framework to compare areas with and without such recorded homicides. The decline in homicides was more pronounced in areas with pre-truce gang exposure.

Following the start of the truce, areas with higher measured gang exposure experienced larger increases in the number of registered firms and formal employment than comparison areas, with the strongest and most stable responses concentrated among small and medium-sized firms. The baseline estimates imply roughly a 3% increase in registered firms and a 6% increase in formal employment. The increase in registered firms is concentrated in non-tradable sectors and driven by both fewer recorded exits and more entries. National household-survey data provide complementary evidence beyond the formal registry: El Salvador's employment-to-population ratio rose temporarily during the truce, a pattern not mirrored in the corresponding Latin American and Caribbean aggregate.

The temporary nature of the truce highlights the fragility of these gains. The agreement lasted approximately two years. As the agreement deteriorated and violence resurged, the earlier gains in registered firms largely reversed. Employment gains were more persistent, but stopped increasing as they had during the truce.

I conduct several robustness checks on the estimated gains during the truce, focusing on the

measurement of gang exposure and differences in pre-truce economic activity. First, I use alternative exposure measures based on gang-related homicides per resident and gang-affiliated prisoners' pre-incarceration residences, including measures of intensity, individual gangs, and the presence of both gangs. I then use matching to compare polygons with more similar pre-truce economic characteristics and separately exclude exposed polygons with the largest firm counts to assess whether a few economically active areas drive the results. I also use log-like outcomes, allow outcome paths to differ across groups defined by initial firm counts, and cluster standard errors by municipality. Across these robustness tests, estimates remain positive and significant for registered firms and employment. Finally, I describe how economic outcomes vary with the size of observed homicide reductions.

To interpret the expansion in business activity, I use a simple framework in which lower violence can increase demand for local goods and services and encourage labor market participation. The concentration of gains in non-tradable sectors and the decline in recorded exits are consistent with stronger customer demand helping existing businesses remain active. Greater demand for their products would also increase firms' demand for labor. Improved consumer confidence and the wage evidence are compatible with this interpretation, without ruling out contributions from labor supply or other adjustment margins.

An additional question is whether the economic gains extended beyond areas with higher measured exposure. Spillovers matter because they may affect the interpretation of the main estimates, but they are also outcomes of independent interest. If neighboring or lower-exposure areas also benefited from improved security, the baseline contrast should be interpreted as a differential-exposure estimate rather than as a pure direct effect. I examine this possibility by adding measures of neighboring exposure and proximity to exposed polygons to the baseline specification. The main estimates remain positive after accounting for neighboring exposure, whose coefficients are also positive. The evidence here is suggestive of positive economic spillovers.

This paper contributes to the growing literature on how organized crime and violence shape economic outcomes, including stock-market valuations and financial conditions (Abadie and Gardeazabal, 2003, 2019; Guidolin and La Ferrara, 2007, 2010), firm performance and survival (Ksoll, Macchiavello and Morjaria, 2010; Camacho and Rodriguez, 2013; Klapper, Richmond and Tran, 2013; Rozo, 2018), and employment and financial stability under criminal capture or infiltration (Fenizia and Saggio, 2024; Mirenda, Mocetti and Rizzica, 2022). It also complements work that quantifies the benefits of crime reduction through perceived valuations (Ludwig and Cook, 2001; Cook,

Jeuland and Ludwig, 2025). Related work also examines how policing and anti-drug enforcement affect the spatial distribution of crime (Draca, Machin and Witt, 2011; Dell, 2015).

More closely related studies examine how violence affects firms, labor market participation, and local economic conditions. Velásquez (2020) finds that increased violence leads self-employed women to reduce their working hours or leave the labor force. Utar (2024) documents declines in blue-collar manufacturing employment consistent with a contraction in labor supply. Brodeur (2018) likewise shows that successful terror attacks reduce local employment and worsen consumer sentiment relative to failed attacks, pointing to the importance of expectations and perceived safety in shaping economic outcomes. Herskowitz et al. (2026) use corporate mobile phone records in Afghanistan to show that violent attacks reduce firms' local presence through both increased exit from affected districts and reduced entry.

This paper builds on that literature by examining how local economic activity responds to a sharp and sustained reduction in lethal violence while criminal organizations remain active. Following these outcomes through the truce and its subsequent breakdown also provides evidence on how durable those gains are when reductions in lethal violence prove temporary.

The analysis covers formal firms across sectors and sizes, extending beyond the listed firms, exporters, and manufacturing plants often studied in this literature. A novel finding is that smaller firms and those in non-tradable sectors, which depend more heavily on local demand, experience the largest gains.

Although the Salvadoran truce was institutionally distinctive, the findings may be relevant to other high-crime settings where small firms and non-tradable sectors play an important economic role. Broader cross-country evidence also points to the relevance of crime for small and medium-sized firms. For instance, Islam (2014) documents a negative correlation between crime and economic activity across firms in 27 developing countries, particularly among small and medium-sized firms.

Because the truce was temporary and its credibility uncertain, the estimates in this paper may understate the economic benefits of more durable and broader security improvements. Firms and workers may have postponed investments and hiring that would have occurred under a permanent decline in violence. The agreement primarily reduced homicides, while other recorded crimes, such as extortion, showed no clear improvement associated with the onset of the truce. The episode illustrates that economic gains can occur alongside such partial improvements in security. Evidence from a later non-aggression pact among the same gangs suggests that extortion may even rise in some

contexts.[1] A more stable and broad-reaching reduction in crime could generate larger increases in economic activity than those estimated here.

The remainder of the paper is structured as follows. Section 2 provides background on violence in El Salvador and the 2012 gang truce. Section 3 describes the data, and Section 4 presents the empirical framework. Section 5 presents the main findings on firm activity and employment, along with supporting credibility checks. Section 6 examines potential margins of adjustment, focusing on firm survival, sectoral composition, wages, and aggregate consumer-confidence indicators. Section 7 uses a simple theoretical framework to interpret the empirical patterns. Section 8 examines spillovers, and Section 9 analyzes the economic reversal as the truce deteriorated and violence rose again. Section 10 concludes.

2 Context

2.1 Violence in El Salvador

El Salvador entered the 2010s with very high levels of violence, much of it linked to gang activity. The country's experience with homicides and extortion was closely tied to the presence of gangs such as MS-13 and Barrio 18. These gangs, formed by Latin American migrants in Los Angeles, expanded after their members' return to El Salvador in a context of weak institutions and limited law enforcement (Arana, 2005). By 2010, crime had become the country's most pressing issue, with 61% of Salvadorans identifying it as their top concern, surpassing economic issues (35%) (Lab, 2010).

Gang members deported from the United States in the mid-1990s established local chapters of MS-13 and Barrio 18. They brought organizational practices, symbols, and identities developed in Los Angeles. The government lacked programs to address gang violence or reintegrate deportees.

These gangs were highly territorial, and their control over neighborhoods restricted residents' mobility. Melnikov, Schmidt-Padilla and Sviatschi (2025) describe how gangs used checkpoints and residential addresses on identification cards to regulate movement across territorial boundaries. Residents crossing these boundaries could face suspicion and the threat of violence. These restrictions could limit access to opportunities outside gang-controlled neighborhoods. Evidence from a later intervention illustrates the relevance of these restrictions. Castro (2026) finds that, following

[1] Brown et al. (2024) studies a 2016 non-aggression pact and documents higher reported extortion, drawing on payment records from a logistics company that distributes goods across El Salvador.

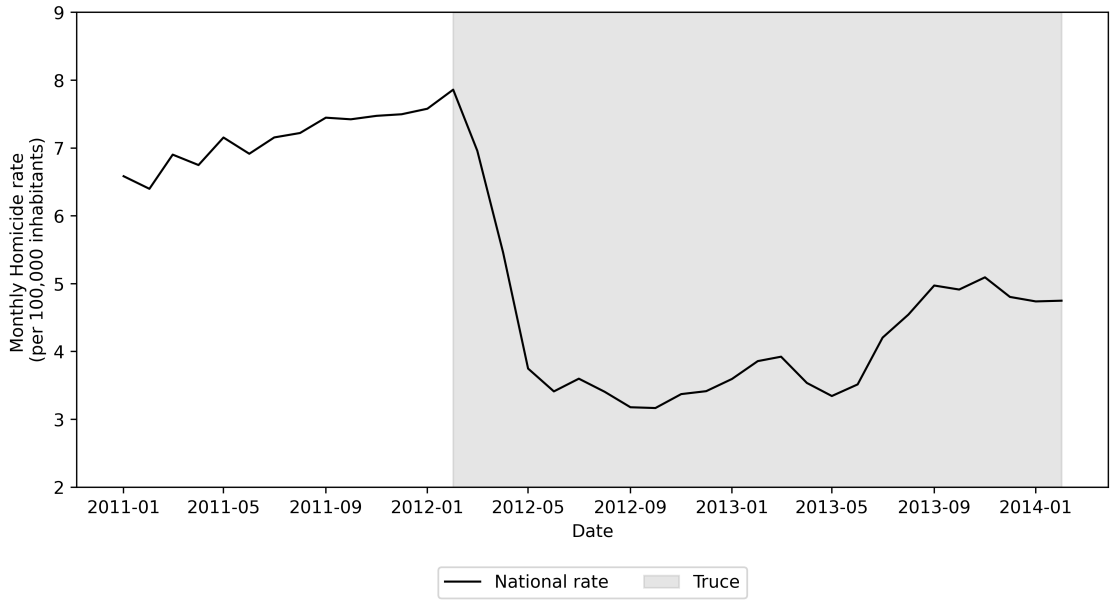
El Salvador's 2022 mass arrests of suspected gang members, previously gang-exposed communities reported larger gains in freedom of movement and greater use of public and community spaces.

Data from the National Police indicate that at least one-third of homicides are attributable to gang-related activities. More broadly, crime and violence impose large economic costs, with estimates suggesting losses between 6.5% and 16% of the country's Gross Domestic Product (GDP) (Jaitman et al., 2017; Peñate et al., 2016). Crime has also been a persistent barrier to investment, with 47% of businesses identifying it as the most critical factor affecting their investment decisions in El Salvador in 2010 (FUSADES, 2010–2020).

2.2 Truce Negotiations

In March 2012, the Salvadoran government facilitated a truce between MS-13 and Barrio 18. Contemporary reporting and subsequent research describe the truce as an initially secret and politically ambiguous process centered on contacts with incarcerated gang leaders (El Faro, 2012; Washington Office on Latin America, 2013; Dudley, 2013; van der Borgh and Savenije, 2019). The truce was not announced ex ante as a public security or economic policy; it became public through press reporting after the homicide decline had begun. The concessions most consistently documented at the onset involved prison transfers, less restrictive prison conditions, and expanded visitation privileges for gang leaders, while the government initially denied having negotiated with gangs (El Faro, 2012; Reuters, 2012; Washington Office on Latin America, 2013; Dudley, 2013; van der Borgh and Savenije, 2019). The historical record does not point to a concurrent public economic, labor-market, or business-formalization policy targeted to areas with greater pre-truce gang exposure. This truce was associated with a roughly 60% reduction in homicides, bringing homicide rates to their lowest levels since the end of the civil war (Washington Office on Latin America, 2013; Dudley, 2013; van der Borgh and Savenije, 2019) (Figure 1).

Figure 1: National Homicide Trends Around the Truce



Notes: This figure plots the national monthly homicide rate before and after the onset of the truce. The series is smoothed using a trailing 3-month moving average. The vertical line indicates the start of the truce period.

Online Appendix Figure B.4 provides additional context by showing national monthly records of homicides, extortion, robbery, and theft around the truce.

The truce is commonly understood to have lasted for approximately 24 months, although its end did not occur on a single sharply defined date. The agreement began to deteriorate as political and institutional support weakened: the Minister of Security associated with the process was removed, and gang leaders were later deprived of the privileges granted during the truce. After this deterioration, the government shifted toward a more confrontational anti-gang strategy, intensifying clashes between security forces and gang members. Homicide rates then surged beyond pre-truce levels, culminating in a rate of 103 per 100,000 inhabitants in 2015, the highest recorded since the civil war.

3 Data

I combine two main administrative data sources for 1,581 polygons covering El Salvador. A polygon is the geographic unit of analysis. These polygons are existing territorial units used for local administrative purposes. They encompass nearby neighborhoods and census areas connected by local roads and sharing services such as schools and polling facilities. I use the same polygon

boundaries throughout the study period. They provide a more granular partition of the country than municipalities. Firm and crime data are aggregated to these polygons and observed over time, yielding a monthly panel from 2010 to 2015.

1. **Social Security Database (ISSS)**: This administrative dataset provides monthly records of formal firms registered with the Salvadoran Social Security Institute. It includes information on firms' primary activity, number of employees, and total payroll.

2. **National Civil Police Records**: This dataset provides monthly records of reported homicides across the country. I use these records to measure local violence and construct the main proxy for measured pre-truce gang exposure, based on pre-truce gang-related homicides.

I also use polygon-level counts of individuals recorded as gang-affiliated, derived from administrative prison records and aggregated by pre-incarceration residence. The homicide records and prison entries used to construct the exposure measures cover 2003–2011, providing a common pre-truce reference period.

I also use aggregate indicators from the Economic Outlook Reports published by FUSADES, a Salvadoran economic and social policy think tank, as complementary contextual evidence. These indicators include consumer confidence, business confidence, and firms' perceived obstacles to investment during the truce period.

3.1 Measuring Pre-Truce Gang Exposure

A central challenge in this setting is that gang presence is not directly observed across locations. While police intelligence provides detailed information on criminal activity, there is no comprehensive measure of the geographic footprint of gangs prior to the truce. To address this, I construct a proxy for pre-truce gang exposure using information on gang-related homicides.

Specifically, I define an indicator equal to one if the police records report gang-related homicides in a polygon during the pre-truce reference period, and zero otherwise.

Gang presence does not always produce a recorded gang-related homicide, so some exposed areas may enter the comparison group. I therefore interpret the indicator as measured gang exposure, rather than a complete map of gang presence. To assess sensitivity to this measure, I also use gang-related homicides per resident and exposure measures based on prisoners' pre-incarceration residences. The intensity measures account for population size, while prisoners' residences provide a separate source of information on gang exposure.

This interpretation has two implications. First, the empirical design compares areas with higher measured exposure to areas with lower or unobserved exposure, rather than fully exposed versus unexposed areas. Second, to the extent that gang-exposed areas are misclassified into the comparison group, the estimated differences may be attenuated if misclassified gang-exposed areas responded more like higher-exposure polygons than like truly lower-exposure areas.

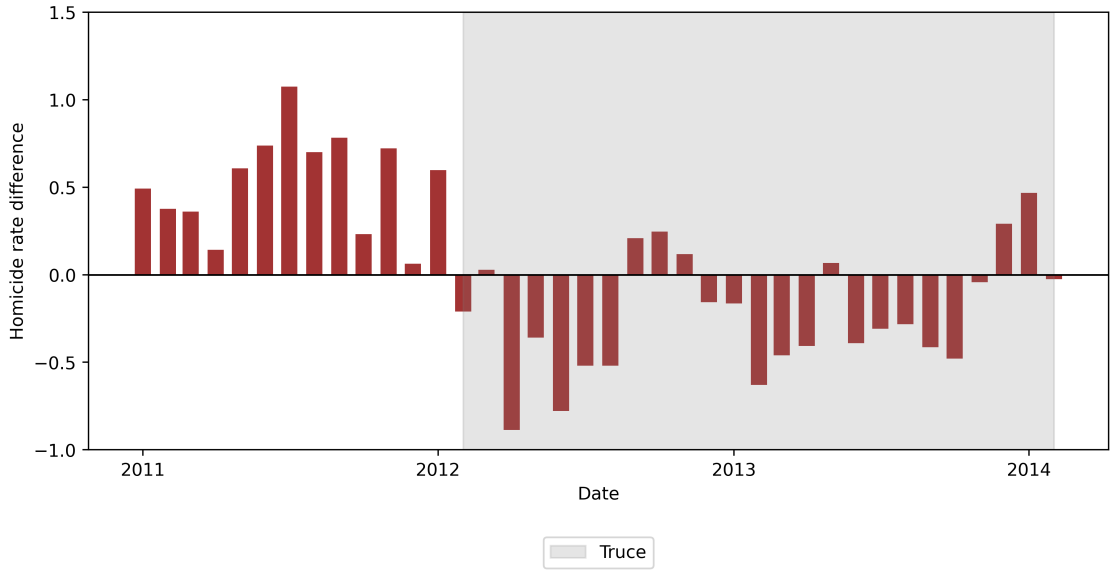
Areas with higher measured exposure differ from lower-exposure areas in baseline economic scale, as shown in Online Appendix Table B.2, panel A. The baseline specification retains the full set of polygons, including areas with little or no pre-truce formal economic activity. Because the empirical design relies on within-polygon changes over time, time-invariant differences in levels are absorbed by unit fixed effects. The natural remaining concern is whether baseline economic scale is associated with differential outcome trends or differential responsiveness to the truce.

I assess these concerns through event studies and comparisons that account for pre-truce economic scale.

3.2 Violence Trends and the Truce

Homicide rates decline in both higher- and lower-exposure areas during the truce. The decline is larger in higher-exposure areas, as shown by the difference in homicide reductions reported in Online Appendix Table B.2, panel B. Figure 2 shows how the homicide-rate gap between the two groups evolves over time.

Figure 2: Difference in Homicide Rates by Pre-Truce Gang Exposure



Notes: This figure plots the monthly difference in homicide rates between polygons with recorded gang-related homicides prior to the truce and those without. The bars summarize how the gap in violence narrows after the onset of the truce, consistent with larger declines in areas with higher measured gang exposure.

4 Empirical Framework

I estimate a difference-in-differences model with polygon and period fixed effects. I use the common timing of the truce together with cross-polygon differences in measured pre-truce gang exposure. Under this approach, $D_i \in \{0, 1\}$ denotes the pre-truce gang-exposure indicator defined above. The estimates capture differential changes by pre-truce gang exposure, rather than the total effect of the truce relative to unaffected areas.

I interpret the estimates as differential local responses to truce-related security conditions, whose most visible and empirically central component was the sharp decline in lethal violence, rather than as the effect of a cleanly isolated policy instrument.

The central comparison assumption is that, absent the truce-induced change in violence, outcomes in high- and low-exposure areas would have followed parallel trends. I assess the plausibility of this assumption using a standard event-study specification:

$$y_{iq} = \alpha_i + \gamma_q + \sum_{k \neq 0} \delta_k \left(D_i \times \mathbf{1}\{q = k\} \right) + \varepsilon_{iq}, \tag{1}$$

Here, q indexes consecutive three-month periods relative to the start of the truce, and y_{iq} denotes the outcome for polygon i in period q. Monthly firm counts and employment are averaged within each polygon-period before estimation. The terms α_i and γ_q denote polygon and period fixed effects, respectively, and D_i is the time-invariant gang-exposure indicator. The omitted period, $q = 0$, is the three-month interval immediately preceding the truce; $q = 1$ is the first truce period. Each coefficient δ_k measures the differential outcome path of higher-exposure polygons relative to lower- or unobserved-exposure polygons, normalized to the omitted pre-truce period.

I also report a complementary specification that uses each polygon's observed homicide reduction during the truce as a continuous measure of treatment intensity. Although homicides fell nationwide, the size of the decline varied substantially across locations. I implement this exercise using the residualized dose-response specification described in Online Appendix Section B.1.2. This specification describes how economic outcomes vary with observed homicide reductions; it is not a separate causal identification strategy.

In the results section, I report estimates based on linear and quadratic specifications of treatment intensity, as well as event-study evidence that describes outcome dynamics across areas with larger and smaller realized homicide reductions.

4.1 Preview of Supporting Checks

I briefly describe the supporting checks that are later reported in the results section; more detailed derivations are provided in the Online Appendix B.1.

For the main specification, I use alternative exposure measures to assess sensitivity to homicide-based classification and population size. Matching and trimming assess whether differences in pre-truce economic scale or a few highly active exposed polygons drive the estimates. I also use the inverse hyperbolic sine transformation to reduce the influence of large outcome values, interact initial firm-count deciles with period effects to allow different time paths, and cluster standard errors by municipality to account for dependence across nearby polygons.

For the descriptive analysis of observed homicide reductions, I report two further checks. First, I examine whether realized homicide reductions are systematically predicted by pre-truce economic conditions. Second, I report placebo exercises that assign larger- versus smaller-decline status in periods before the truce and consider alternative intensity thresholds. These checks help characterize whether the descriptive gradient is related to pre-truce economic growth measures or appears similarly when the homicide-reduction measure is assigned to placebo periods.

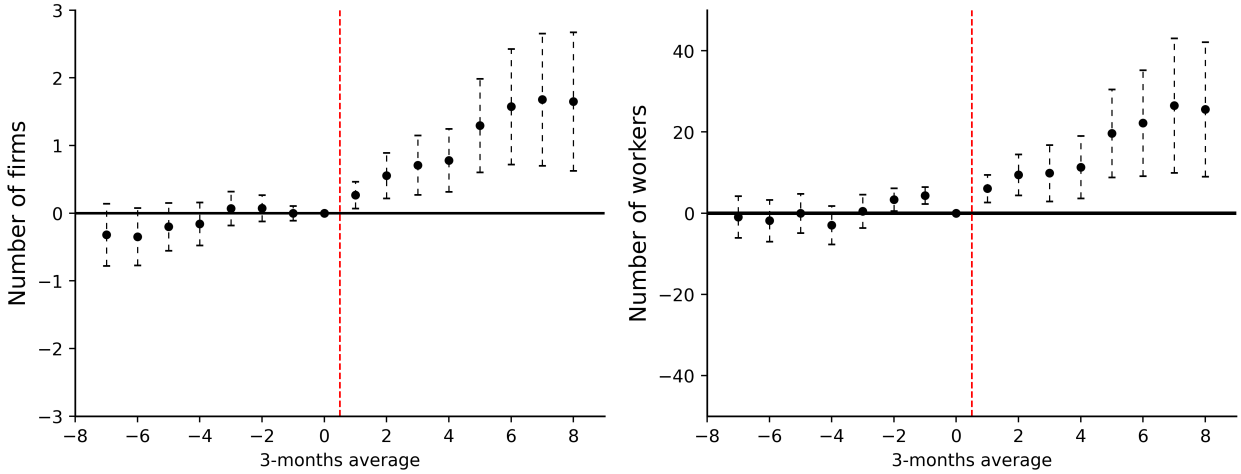
5 Main Results

This section presents the main empirical findings on the economic effects of the truce. I first compare changes in firm counts and employment across areas with different levels of pre-truce gang exposure. I then assess the robustness of these patterns to alternative comparison strategies, and finally examine how outcomes vary with the magnitude of violence reductions across locations.

5.1 Main Effects of the Truce

Figure 3 presents event-study estimates for the number of firms and workers across areas with different levels of pre-truce gang exposure, focusing on small and medium firms (bottom 95%). The corresponding results for the full sample are reported in Appendix Figure A.1. Before the truce, the estimates do not show a sharp break around the onset of the truce. After the onset of the truce, higher-exposure areas exhibit increases in the number of registered firms relative to comparison areas, accompanied by a sustained expansion in employment. Taken together, the event-study pattern is consistent with a relative improvement in local economic activity in areas with greater pre-truce gang exposure.

Figure 3: Event-Study Estimates for Firms and Workers, Excluding the Top 5% Largest Firms



Notes: This figure displays event-study estimates from Equation (1)*. The X-axis consolidates data into three-month intervals, covering 24 months before and after the onset of the truce. The omitted category is the three-month interval immediately preceding the truce. All regressions include unit and time fixed effects, and the sample excludes the top 5% of firms by size. Vertical bars indicate 95% confidence intervals, with standard errors clustered at the geographic unit level.*

Table 1 reports the corresponding difference-in-differences estimates. For small and medium firms (bottom 95%), the estimates imply an increase of about 1.2 registered firms per polygon (roughly 3 percent relative to the pre-truce mean) and about 16 workers (roughly 6 percent), both statistically significant. By contrast, the estimates for the top 5% of firms are statistically indistinguishable from zero. When all firms are pooled together, the estimates remain positive: the number of firms rises by about 1.17 per polygon (roughly 2.9 percent, statistically significant), and employment increases by about 20 workers (roughly 2.7 percent, marginally significant). The strongest evidence of gains comes from the bottom 95% of firms.

Table 1: Main Difference-in-Differences Estimates for Firms and Employment

Variables	**Number of firms**			**Workers**		
	Bottom 95%	Top 5%	All	Bottom 95%	Top 5%	All
Treatment	1.175***	-0.008	1.167***	16.027***	3.988	20.014*
	(0.366)	(0.016)	(0.369)	(5.747)	(6.478)	(10.426)
Mean	38.314	2.300	40.614	267.543	463.082	730.624
R-squared	0.9986	0.9993	0.9988	0.9958	0.9967	0.9973
Obs.	25296	25296	25296	25296	25296	25296
Clusters	1581	1581	1581	1581	1581	1581

Notes: This table reports difference-in-differences estimates of differential post-truce changes in the number of firms and employment across areas with higher versus lower measured pre-truce gang exposure. Estimates are obtained from a two-way fixed effects (TWFE) specification analogous to Equation (1)*. Results are shown for all firms and separately for small and medium firms (bottom 95%) and large firms (top 5%). All regressions include unit and time fixed effects. Standard errors, clustered at the geographic unit, are shown in parentheses.* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

I next examine whether the results depend on the binary characterization of gang exposure or on the scale of the outcomes. Appendix Table A.1, panels A–C, reports these specifications for registered firms and formal employment in the bottom-95% sample, with corresponding event studies in Appendix Figure A.2. First, I replace the binary exposure indicator with a continuous measure based on pre-truce gang-related homicides per 1,000 residents, transformed using the inverse hyperbolic sine and standardized across polygons. This measure retains variation in recorded homicide exposure that the binary indicator discards while accounting for population size. The estimates remain positive and statistically significant for both outcomes.

Second, I re-estimate the main binary-exposure specification using the inverse hyperbolic sine of the outcomes, a log-like transformation that accommodates zeros. By compressing large values, this

specification assesses whether the results persist when changes are measured on a scale closer to proportional rather than absolute differences. I then interact pre-truce firm-count deciles with three-month period indicators, allowing outcomes to follow different time paths across groups defined by average pre-truce firm counts. Separately, I cluster standard errors by municipality to allow for dependence across polygons within the same municipality, and I report a specification combining both adjustments. Estimates remain positive and statistically significant for both outcomes across these specifications.

Third, I construct alternative exposure measures using the pre-incarceration residences of individuals recorded as gang-affiliated who entered prison during the pre-truce reference period. This source provides a residential proxy for pre-truce gang exposure that does not rely on the location of recorded homicides. I consider an indicator for polygons with recorded residents from each of the two gangs, MS-13 and Barrio 18, a population-adjusted intensity measure, and separate indicators for each gang. Estimates are positive across all four definitions for both outcomes.

Taken together, these findings indicate that areas with greater measured pre-truce gang exposure experienced meaningful relative improvements in local economic activity after the truce. The effects are concentrated among small and medium firms, where both firm counts and employment increase, while large firms show little detectable adjustment.

5.2 Addressing Comparability

A central concern is that initial economic scale may influence how polygons respond to improvements in security. Polygons with higher measured gang exposure tend to have more firms and workers before the truce. The baseline comparison could therefore reflect both differences in exposure to the decline in violence and differences in responsiveness associated with pre-truce economic activity.

To address this concern, I implement a matched-sample design that restricts comparisons to areas with similar levels of pre-truce economic activity. Specifically, I match higher-exposure polygons to lower- or unobserved-exposure polygons based on the number of firms observed prior to the truce. This variable provides a direct proxy for the local scale of economic activity, capturing the size of the local business base. As shown in Online Appendix Table B.2, panel A, matching on this dimension also improves balance across several other baseline characteristics, although I do not interpret the matched sample as fully balanced across all pre-truce characteristics.

I denote these specifications as $NN(k)$, where k is the number of nearest neighbors used to

construct the matched comparison group. The preferred matched specification, $NN(5)$, uses five-nearest-neighbor caliper matching on pre-truce firm counts and matches without replacement. This is a restrictive design: each comparison polygon can enter at most once, so the estimates do not rely heavily on repeated use of a small number of comparison areas.

Online Appendix Table B.2, panel A, documents these balance improvements. Differences in pre-truce firm counts fall substantially, and the matched sample is also closer in employment, wages, and opening and closure rates. As is typical with common-support restrictions, the matched estimates pertain to treated polygons for which close comparison areas are available.

Appendix Figure A.2, panels M–N, and Appendix Table A.1, panel D, report the results for the preferred $NN(5)$ specification. The estimates remain positive and statistically significant, consistent with larger post-truce increases in areas with higher measured gang exposure, even among polygons with more similar pre-truce economic activity. The matched event studies show no clear systematic divergence before the truce and positive post-truce coefficients for both outcomes. Although employment is less precisely estimated, its post-truce response becomes more pronounced during the second year.

Under the preferred $NN(5)$ specification, the estimates imply increases of approximately 4 percent in registered firms and 3.3 percent in formal employment, relative to the respective pre-truce means among matched exposed polygons. The firm-count response is similar in magnitude to the baseline estimate. For employment, the corresponding baseline estimate is 6.0 percent; the difference between these coefficient-to-mean ratios is not statistically significant.

I also consider an $NN(15)$ specification with more matched neighbors, a tighter caliper, and limited reuse of comparison polygons. This specification improves balance in pre-truce firm counts and other baseline characteristics (Online Appendix Table B.2, panel A). The relative magnitudes are similar to those under $NN(5)$ (Appendix Table A.1, panel D). Firm-count estimates remain statistically significant, while employment estimates are less precise.

I also examine whether exposed polygons with the highest pre-truce firm counts drive the estimates. I exclude the 5 and 15 percent of exposed polygons with the highest average pre-truce firm counts, while retaining all comparison polygons. The estimates remain positive and statistically significant for both outcomes, corresponding to 3.4–4.5 percent of the pre-truce mean for registered firms and 4.1–5.2 percent for formal employment (Appendix Table A.1, panel D). Appendix Figure A.2, panels Q–T, presents the corresponding event studies.

5.3 Continuous Variation in Violence Reductions

To better describe the magnitude of the economic response, I use continuous variation in the decline in violence across areas. Specifically, I construct a standardized measure of the reduction in homicides following the truce, allowing me to characterize how local economic outcomes vary with a one standard deviation larger decline in violence. The technical details of this complementary specification are described in Online Appendix Section B.1.2. This exercise is descriptive.

Table A.2 reports the corresponding regression estimates. A one standard deviation decrease in violence is associated with an increase of about 4.6 percent in the number of firms and about 8.4–8.5 percent in employment after one year. The implied associations grow over the two-year horizon, reaching close to 9 percent for firms and roughly 17 percent for employment. The corresponding event-study estimates, shown in Appendix Figure A.3, do not indicate differential pre-truce movements under the linear continuous specification.

A key concern with this complementary exercise is that areas experiencing larger realized declines in violence may also differ in underlying growth dynamics. To assess this, I group areas according to the magnitude of violence reductions and estimate event-study specifications for each group. Figure A.4 shows broadly similar pre-truce movements across intensity groups and a clearer post-truce gradient, with larger improvements in areas experiencing larger realized declines in violence.

I further examine whether the magnitude of the decline in violence is systematically related to pre-existing economic conditions. Table A.3 shows that population size and pre-truce violence are the most consistent predictors of the size of the observed homicide decline. By contrast, baseline firm counts, employment, wages, and pre-truce growth do not consistently predict the size of the decline once these demographic and violence controls are included. This helps address the concern that realized homicide reductions are mainly selecting polygons with stronger pre-truce economic growth dynamics.

Finally, I use placebo exercises to assess whether the homicide-decline classification itself could generate the observed economic patterns, for example by selecting areas experiencing mean reversion in violence. I apply the same classification procedure at placebo dates preceding the intervention and jointly estimate the placebo and actual-truce specifications, as described in Online Appendix Section B.2.1. These checks also consider alternative intensity thresholds that more closely mirror the dose-response exercise. Under the baseline classification, the average placebo estimates are

small and statistically indistinguishable from zero for both outcomes, while the incremental truce estimates remain positive and close in magnitude to the corresponding difference-in-differences estimates (Online Appendix Table B.5).

Taken together, these results show a positive gradient between realized homicide reductions and local economic activity. The supporting checks indicate that similar patterns are not present in placebo periods and that the realized reductions are not strongly predicted by pre-truce economic growth measures.

6 Mechanisms: Firm Survival, Local Demand, and Labor Markets

The previous section documents gains in registered firms and employment, especially among small and medium-sized firms. Did employment rise because more people were willing to work, because firms wanted to hire more workers, or both? Higher employment alone does not distinguish between these explanations: safer conditions can increase labor supply by making work less costly, but they can also stimulate spending and increase firms' demand for labor.

To assess these explanations, I examine firm entry and exit, sectoral activity, wages, and confidence indicators. Entry and exit show whether the increase in registered firms reflects new registrations, fewer recorded exits, or both, while sectoral patterns indicate whether gains are concentrated in activities tied to local demand. Wage responses help assess the relative roles of labor supply and labor demand, and confidence indicators provide complementary evidence on household expectations and firms' perceived constraints. Taken together, the evidence suggests that increased demand for labor was an important margin of adjustment, without ruling out contributions from labor supply or other channels.

6.1 Firm Survival and Entry

I first examine entries into and exits from the firm registry. Entry may reflect either business creation or formalization, while exit indicates that a firm is no longer observed in the registry through the end of the sample. Table 2 reports difference-in-differences estimates for business closures and openings. Within each outcome, *Main* uses pre-truce gang exposure. *Split* compares the top and bottom halves of the homicide-reduction distribution during the truce, while the *Top* specifications retain progressively smaller subsets of the high-decline group.

The estimates show that areas with higher measured pre-truce gang exposure experienced both

fewer closures and more openings after the truce, relative to areas with lower or unobserved measured exposure. Closures fall by about 10 percent, while openings increase by roughly 8 percent; both estimates are statistically significant. The reduction in recorded exits is particularly informative because it points to a survival margin among existing businesses, rather than an expansion operating only through new entry. This pattern is consistent with stronger local demand helping incumbent businesses remain active.

The estimates retain the same signs when I compare polygons with larger and smaller observed homicide declines and progressively restrict the high-decline group. The decline in recorded exits is therefore also evident under classifications based on the size of the observed violence reduction.

Table 2: Business Closures and Openings by Measured Gang Exposure

Variables	**Closing business**					**Opening business**				
	Main	Split	Top 75	Top 50	Top 25	Main	Split	Top 75	Top 50	Top 25
Estimate	-0.082**	-0.041**	-0.059***	-0.076**	-0.108**	0.074**	0.035*	0.040*	0.056*	0.065
	(0.036)	(0.018)	(0.022)	(0.030)	(0.051)	(0.035)	(0.019)	(0.023)	(0.030)	(0.048)
Per. change	-10.2%	-9.7%	-11.4%	-12.3%	-13.5%	8.1%	5.7%	5.4%	6.4%	6.7%
Mean	0.805	0.422	0.519	0.618	0.799	0.911	0.610	0.738	0.876	0.974
R-squared	0.7826	0.7825	0.7905	0.8056	0.8324	0.8321	0.8320	0.8408	0.8549	0.8777
Obs.	25296	25296	22192	19104	16016	25296	25296	22192	19104	16016
Clusters	1581	1581	1387	1194	1001	1581	1581	1387	1194	1001

Notes: This table reports difference-in-differences estimates for business closures and openings among small and medium firms (bottom 95%). The first five result columns report closing outcomes; the next five report opening outcomes. Within each outcome, Main *uses the main exposure indicator, which equals one for polygons with at least one observed gang-related homicide before the truce and zero for polygons with no such observed homicide. The remaining specifications use complementary classifications based on realized homicide reductions during the truce.* Split *compares polygons in the top half of the homicide-reduction distribution to polygons in the bottom half.* Top 75*,* Top 50*, and* Top 25 *further restrict the high-decline group within that top half, retaining only polygons above the 75th, 50th, and 25th percentiles of realized homicide reductions among high-decline polygons, respectively. Percentage changes are computed relative to the pre-truce mean reported in the table. All regressions include unit and time fixed effects. Standard errors, clustered at the geographic unit, are shown in parentheses.* $^{***}p < 0.01$*,* $^{**}p < 0.05$*,* $^{*}p < 0.1$*.*

One potential concern is that the increase in registered firms could partly reflect transitions from informal to formal activity, since the administrative data only cover formal-sector firms. This margin remains possible. However, it is unlikely to account for the full pattern. A related concern is that a contemporaneous registration reform, rather than the truce, could have encouraged firms to enter

the formal registry. The World Bank's Doing Business reform history records no business-registration reform in El Salvador in the 2011–2014 editions; the last recorded reform on this margin appears in the 2009 edition (World Bank, 2013). This makes it unlikely that a contemporaneous formalization or registration reform explains the timing of the firm response. Aggregate employment data from the World Development Indicators provide a further perspective because the employment-to-population ratio includes both formal and informal employment. This ratio rose during the truce before returning to pre-truce levels (Online Appendix Figure B.6 and Table B.1). A shift from informal to formal employment alone would not increase this measure, so the national pattern provides complementary evidence consistent with employment gains beyond reclassification.

6.2 Sectoral Composition and Local Demand

I next examine whether the response is concentrated in sectors that are more dependent on local market conditions. Following Knight and Johnson (1997), I classify sectors as tradable or non-tradable based on their exposure to broader markets. This comparison is useful because a response driven by stronger demand for local goods and services should appear more strongly in non-tradable activities than in sectors whose demand is less tied to nearby consumers.

Table 3 reports the sectoral decomposition. The pooled sector columns estimate a single stacked regression with polygon-sector and time fixed effects, allowing the post-truce response to differ directly between tradable and non-tradable activity. The estimates show that the positive aggregate effect is concentrated in non-tradable sectors. Tradable activity declines modestly after the truce, by about 0.2 firms and 2.2 workers per polygon, while non-tradable activity increases by about 1.4 firms and 18.2 workers.

Table 3: Difference-in-Differences Estimates by Tradability

Specification	Number of firms		Workers	
	Baseline DID	Pooled sector	Baseline DID	Pooled sector
Baseline DID	1.175***		16.027***	
	(0.366)		(5.747)	
Tradable		-0.210***		-2.182***
		(0.063)		(0.724)
Non-tradable		1.384***		18.209***
		(0.334)		(5.389)
Mean	38.314	38.314	267.543	267.543
R-squared	0.9986	0.9986	0.9958	0.9956
Obs.	25296	50592	25296	50592
Clusters	1581	1581	1581	1581

Notes: This table reports difference-in-differences estimates for the number of firms and workers. The pooled sector columns estimate a stacked sector-level specification with polygon-sector fixed effects and time fixed effects. Tradable and non-tradable sectors are classified following Knight and Johnson (1997). Standard errors, clustered at the geographic unit, are shown in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The more disaggregated sector patterns point in the same direction. Online Appendix Figure B.9 shows that the largest absolute gains are concentrated in activities closely tied to local commerce and services, especially **G** (*Commerce*), **K** (*Real Estate, Renting, and Business Activities*), and **I** (*Transport, Storage, and Communications*). Other locally oriented activities, such as **H** (*Hotels and Restaurants*), **O** (*Other Community, Social, and Personal Services*), and **F** (*Construction*), also display positive responses, particularly when scaled relative to their smaller pre-truce baselines. This composition strengthens the interpretation that the response is linked to local market activity rather than to a broad expansion in externally oriented production.

The sectoral evidence therefore points to locally oriented, non-tradable activity as the main source of the aggregate gains.

6.3 Wages and Local Labor Market Adjustment

Wages provide a useful, but more limited, way to assess whether the employment gains were accompanied by changes in local labor-market conditions. Safer conditions could increase labor supply by making work less costly and increase firms' demand for labor by encouraging spending on local goods and services. The wage evidence helps assess which force appears more salient, although the estimates are imprecise. Because wage measures are noisy in small geographic units, I restrict

the sample to polygons with more than 10 total employees and code wages as missing when the corresponding worker denominator is zero.

Appendix Table A.4 reports estimates for monthly wages per worker. Under the main measured-exposure specification, the coefficients are positive for the bottom 95%, top 5%, and all-firms samples, but they are not statistically significant. Estimates are larger when I compare polygons in the top half of the observed homicide-reduction distribution during the truce with those in the bottom half: wages in the bottom 95% increase by about 2.1 percent and wages for all firms increase by about 1.9 percent, with the former statistically significant and the latter marginally significant.

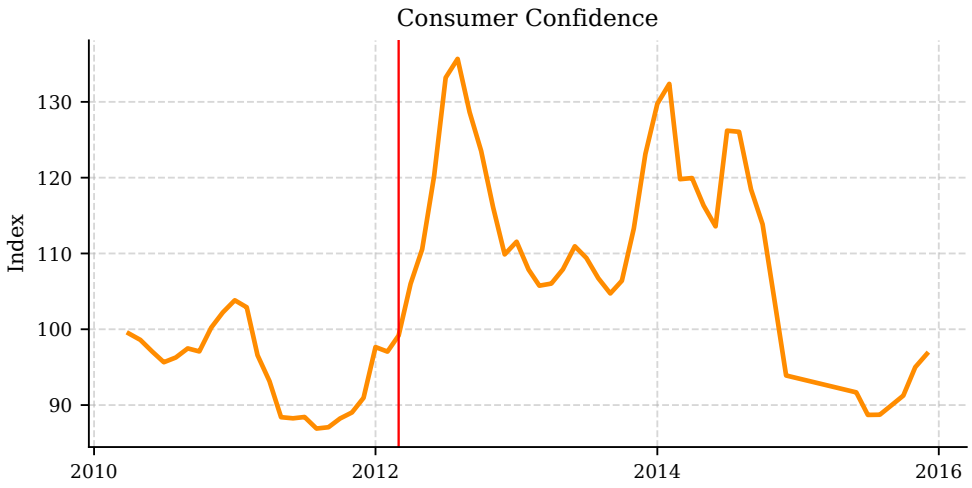
I interpret this evidence cautiously. The wage estimates are less precise than the firm-count, employment, and closure results, and they are not strong enough to establish a separate labor-market channel. At the same time, the estimates do not point to wage declines, which would be more natural if the employment gains reflected only an outward shift in labor supply. They are therefore best read as supportive evidence that the employment gains are compatible with demand-side pressure being an important margin, while leaving substantial uncertainty about the extent of wage adjustment.

6.4 Confidence and Perceived Crime Constraints

I also examine aggregate indicators from FUSADES Economic Outlook Reports to describe changes in perceptions and expectations during the truce. Figure 4 plots consumer confidence, based on households' assessments of current conditions, major purchases, and economic expectations. Appendix Figure A.5 reports firms' perceived investment obstacles and business confidence.

The most informative pattern is the timing of consumer confidence. The index rises sharply around the onset of the truce. This timing is consistent with a setting in which improved security made households more willing to move, purchase, and transact locally, making demand-side responses more plausible. Business confidence does not show a similarly timed change around the onset of the truce, so I place less weight on this series.

Figure 4: Consumer Confidence Around the Truce



Notes: The figure plots the consumer confidence index from FUSADES Economic Outlook Reports as a four-month moving average. The red vertical line marks the start of the truce in March 2012.

LAPOP survey evidence also shows a 12-percentage-point decline between 2010 and 2012 in the share of respondents identifying crime and insecurity as the country's most pressing problem, coinciding with the year the truce began (Online Appendix Figure B.12, panel B).

The perceived investment constraints point in the same direction from the firm side. The share of firms citing crime as the main obstacle to investment fell from roughly 42 percent before the truce to about 27 percent during the truce, before returning to approximately its pre-truce level afterward. Over the same period, uncertainty became relatively more salient as crime became less frequently cited as the main obstacle. These shares describe how firms' reported investment concerns changed during the truce. Together with the confidence indicators, these changes provide aggregate evidence consistent with the patterns in registered firm activity. Consumers became more optimistic and fewer firms cited crime as their main investment obstacle during the period when exposed areas experienced fewer recorded exits and stronger activity in sectors serving nearby consumers.

The clearest patterns are fewer firm closures and stronger activity in locally oriented sectors. Wage estimates and confidence indicators are less definitive but move in a direction consistent with stronger local activity. Taken together, the evidence suggests that stronger local demand was an important margin behind the main effects, even if labor-supply responses also contributed to employment growth.

7 Discussion

A simple theoretical framework helps organize two remaining interpretive questions (formal details in Appendix B.4). First, when safer conditions can increase both labor supply and local demand, what makes the demand-side margin more likely to dominate? Second, when economic activity rises in more-exposed areas, should this be interpreted as net expansion or as spatial reallocation from less affected areas?

In the model, the demand response depends on spending in the market affected by crime, not simply on the number of workers who live there. Lower fear and transaction costs increase demand for goods sold in that market and, in turn, firms' demand for labor. This response can outweigh the increase in labor supply when the affected market accounts for enough potential spending and demand is sufficiently sensitive to crime. The mechanism may be relevant for the urban and commercial areas affected by gangs in El Salvador.

This distinction helps interpret the mechanism results. If the decline in violence reduced local fear and transaction costs in economically meaningful local markets, the expected pattern is closer to what appears in the data: fewer firm exits, stronger activity in sectors tied to nearby consumers, and no evidence of wage declines.

The second issue is reallocation. In the framework, a decline in crime increases aggregate employment because safer conditions expand both the willingness to work and the demand for local goods. At the same time, local gains can include spatial reallocation: activity may shift from less affected areas toward areas where security improves. The extent of this reallocation depends on which force dominates. If the demand-side response is strong enough to make the newly safer market relatively more attractive, some workers, customers, or business activity may move toward the more-exposed locations. If the labor-supply response is stronger, aggregate employment can rise with less displacement from other areas.

The empirical evidence provides a complementary perspective. The temporary rise in national employment-to-population is consistent with broader employment gains. The decline in recorded exits suggests that the gains were not simply a matter of attracting businesses from elsewhere: businesses already present in exposed areas were also less likely to leave the registry.

I next examine whether economic gains also appeared in neighboring polygons.

8 Additional Evidence on Spillovers

Spillovers matter for two reasons. First, neighboring areas may also benefit from improved security through shared demand, mobility, commuting, or commercial linkages. Second, indirect exposure can affect interpretation if neighboring polygons that respond to the truce remain in the comparison group. This does not invalidate the measured-exposure comparison, but it clarifies the estimand: the baseline estimates capture differential gains by measured exposure, not the total effect of the truce relative to unaffected areas.

To assess this possibility, I augment the baseline specification with measures of neighboring exposure. The first measure indicates whether a polygon has spatial contact with a polygon that had measured pre-truce gang exposure. The second measure uses proximity to the nearest exposed polygon, with higher values corresponding to closer locations. These exercises should be interpreted cautiously. The exposure proxy is measured with error, and spatial contact or proximity captures only one form of interference; spillovers may also operate through commuting, commercial linkages, transport routes, firm relocation, or broader market connections.

Appendix Table A.5 reports the corresponding estimates. The direct-exposure coefficient remains positive after accounting for neighboring exposure. The coefficients on neighboring exposure and proximity are also positive, suggesting that economic gains may have extended beyond directly exposed polygons. This pattern is consistent with local improvements spreading through nearby markets, although the estimates are best read as suggestive evidence rather than a separate source of identification.

I also broaden the exposure definition to include polygons that border directly exposed areas (Appendix Table A.6). The estimates remain positive for both employment and firm counts, suggesting that the main findings are robust to this broader exposure definition.

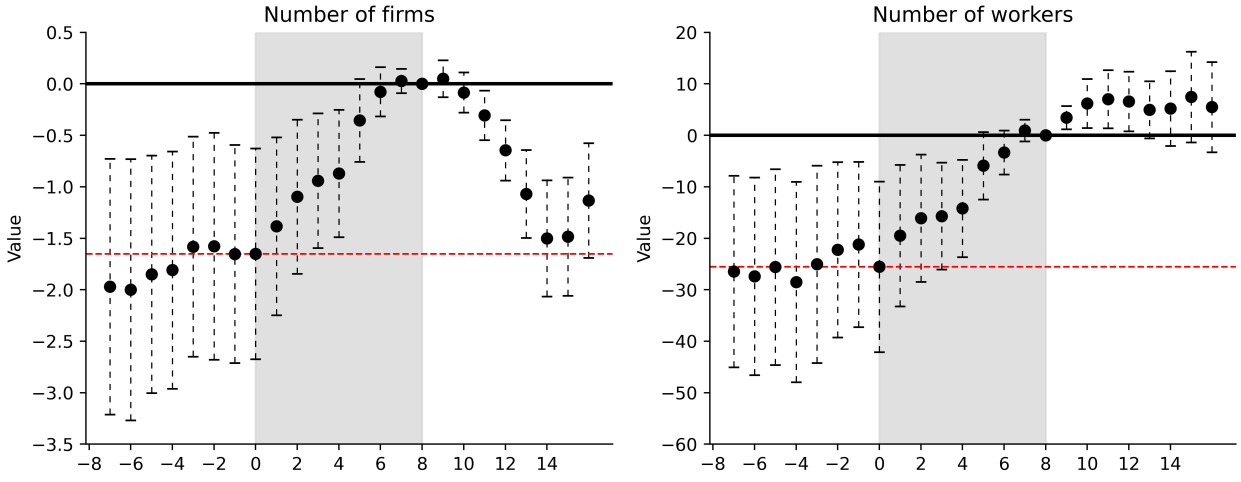
9 Aftermath of the Truce: Rising Crime and Economic Reversal

Shortly after the truce dissolved, the national homicide rate began to rise steadily, eventually surpassing its pre-truce level. Appendix Figure A.6 shows this reversal. By 2015, El Salvador recorded the most violent year in its recent history, with the national homicide rate exceeding 100 per 100,000 inhabitants. This surge in violence coincided with a sharp escalation in the government's anti-gang efforts, often referred to as *mano dura* ("iron-fist") policies, which emphasized aggressive policing.

These confrontational tactics intensified conflict between security forces and gang networks, likely fueling further instability and reversing many of the temporary security gains achieved during the truce.

In tandem with rising crime, the economic indicators that had improved during the truce began to reverse. Figure 5 shows that the relative gains in firm counts largely reversed by the end of 2015. Employment gains stopped increasing as they had during the truce, but remained above the pre-truce reference for much of the post-truce period. Interestingly, both higher- and lower-exposure areas experienced increases in crime after the ceasefire collapsed, yet higher-exposure areas exhibited a sharper decline in active firm counts as violence rose. One possible explanation is that these areas had shown larger gains during the truce—consistent with a substantial "peace dividend"—and were therefore more vulnerable to resurgent insecurity. This suggests that firms in higher-exposure areas may have adjusted their operations under the assumption of sustained improvements in security, making them more susceptible to instability when violence returned.

Figure 5: Trends in the Number of Firms and Workers Post-Truce



Notes: This figure presents event-study estimates from a specification analogous to Equation (1)*, redefining the event time to center around the end of the gang truce. The estimates compare areas with higher measured pre-truce gang exposure to areas with lower or unobserved measured exposure. The dependent variables are the number of firms (left panel) and number of workers (right panel). The red dotted horizontal line indicates the estimated pre-truce reference level, which serves as a benchmark for assessing how much of the truce-period gain remains after the truce ends. The shaded gray area highlights the duration of the truce. Time is grouped into three-month intervals. All regressions include unit and time fixed effects. Vertical bars represent 95% confidence intervals, with standard errors clustered at the geographic unit level.*

This reversal should be interpreted descriptively, since the end of the truce coincided with broader policy changes, including the escalation of *mano dura* strategies. Nevertheless, the reversal of firm-count gains highlights their vulnerability to renewed insecurity.

Additional context comes from a later, more limited decline in homicides in early 2016, following a non-aggression pact among gangs widely understood as a response to the government's intensifying security measures (Gagne, 2017). Although the agreement lowered the national homicide rate, crime levels remained close to their pre-truce average and public sentiment did not improve: consumer confidence and perceptions of safety showed no meaningful gains (see Online Appendix Figure B.12). Using data from a large distributor during this episode, Brown et al. (2024) document an increase in extortion payments that was passed through to retailers and consumers, without significant changes in the distributor's delivery activity.

10 Conclusion

This paper studies how a sharp reduction in homicides during the 2012 Salvadoran gang truce relates to registered firm activity and formal employment. The truce produced a large decline in homicides, while areas differed in their measured pre-truce exposure to gang activity. I use this variation to compare post-truce changes in areas with higher measured gang exposure to changes in areas with lower or unobserved measured exposure.

The results show meaningful gains in local economic activity following the truce. Areas with higher measured pre-truce gang exposure experienced larger post-truce increases in the number of firms and in employment, with the strongest and most stable responses concentrated among small and medium-sized establishments.

Areas with higher measured exposure experienced fewer recorded exits and more entries, pointing to business survival as an important margin of adjustment. Sectoral results show that the gains are concentrated in non-tradable activities serving local markets. Wage estimates and aggregate consumer confidence indicators are less definitive, but they move in a direction consistent with stronger local demand.

The analysis also suggests that these gains may have extended beyond narrowly measured exposure boundaries. Neighboring areas show suggestive evidence of improved outcomes, and specifications that group polygons with measured pre-truce exposure and adjacent polygons produce similar positive patterns. At the same time, the gains appear fragile. When violence rose again after

the collapse of the truce, firm-count gains reversed more clearly than employment gains, suggesting that the businesses most responsive to safer conditions were also vulnerable to renewed insecurity.

Taken together, the findings suggest that reductions in lethal violence can support registered business activity and formal employment even while criminal organizations remain active. The gains are concentrated among smaller, locally oriented establishments, and their subsequent weakening underscores the importance of sustained improvements in security.

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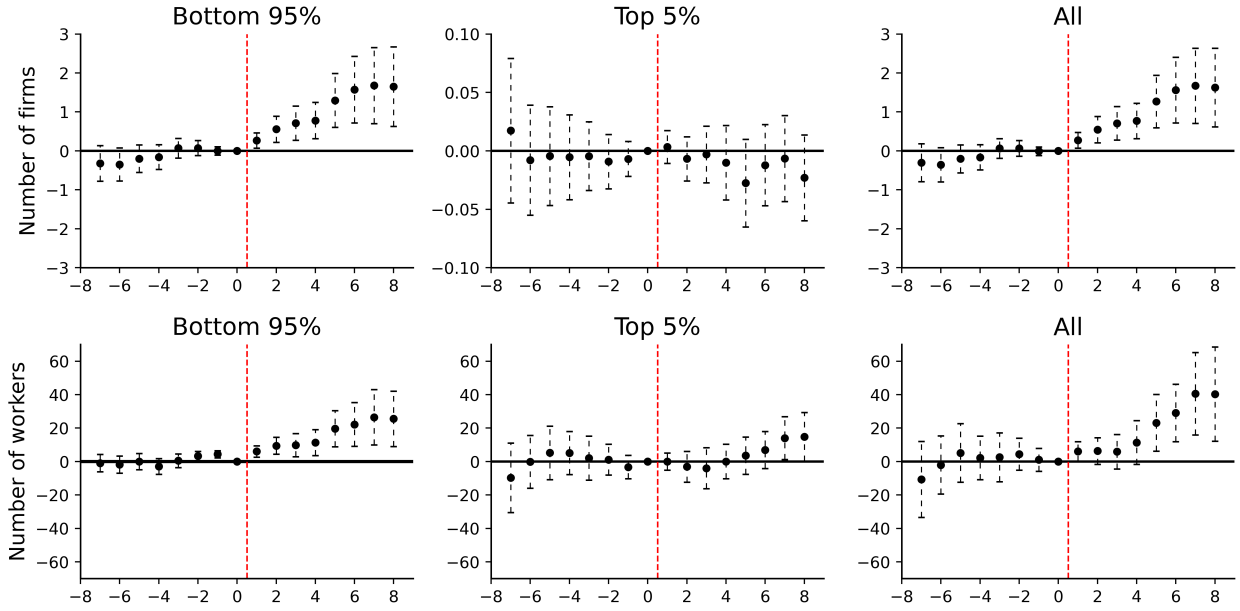
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A Appendix

A.1 Figures and Tables

Figure A.1: Event-Study Estimates for Firms and Workers, Full Sample



Notes: This figure displays event-study estimates from Equation (1) *for the full sample of firms. The X-axis consolidates data into three-month intervals, spanning 24 months before and after the truce. All regressions include unit and time fixed effects. Vertical bars indicate 95% confidence intervals, with standard errors clustered at the geographic unit level.*

Table A.1: Alternative exposure definitions and functional form

Registered firms and formal workers, bottom 95%

	Registered firms					Formal workers				
Exposure / specification	Coef.	SE	% mean	p	Obs.	Coef.	SE	% mean	p	Obs.
A. Benchmark and homicide-based exposure										
Main treatment: bottom 95%	1.175***	0.366	3.07	0.001	25,296	16.027***	5.747	5.99	0.005	25,296
Gang-homicide intensity[b]	0.376***	0.119	0.98	0.002	25,296	5.492***	1.841	2.05	0.003	25,296
B. Main treatment: asinh outcomes, scale controls, and inference										
Asinh only[a]	0.037***	0.012	3.81	0.002	25,296	0.048*	0.025	4.87	0.055	25,296
Asinh + decile effects[a]	0.047***	0.014	4.77	< 0.001	25,296	0.057**	0.028	5.88	0.041	25,296
Asinh + municipality clustering[a]	0.037***	0.014	3.81	0.007	25,296	0.048**	0.023	4.87	0.036	25,296
Asinh + both adjustments[a]	0.047***	0.014	4.77	< 0.001	25,296	0.057**	0.025	5.88	0.021	25,296
C. Prisoner-residence exposure (imputation with initial-firm-group effects)										
Prisoners: gangs present	0.705**	0.295	2.15	0.017	25,296	9.733**	4.193	4.26	0.020	25,296
Prisoners: intensity[b]	0.396***	0.137	1.89	0.004	25,296	3.752**	1.693	2.63	0.027	25,296
Prisoners: MS13	0.656***	0.212	2.86	0.002	25,296	5.769**	2.646	3.69	0.029	25,296
Prisoners: B18	0.456*	0.242	1.68	0.060	25,296	7.023**	3.335	3.73	0.035	25,296
D. Matched and trimmed samples (main treatment)										
NN(5), caliper 0.35, reuse 0	0.467***	0.155	4.01	0.003	20,448	2.333*	1.341	3.27	0.082	20,448
NN(15), caliper 0.25, reuse 4	0.561***	0.184	4.26	0.002	16,560	2.766*	1.640	3.35	0.092	16,560
Treated-only trimming: 5%	0.798***	0.200	3.37	< 0.001	25,056	7.815***	1.794	5.16	< 0.001	25,056
Treated-only trimming: 15%	0.675***	0.177	4.55	< 0.001	24,576	3.740**	1.543	4.15	0.015	24,576

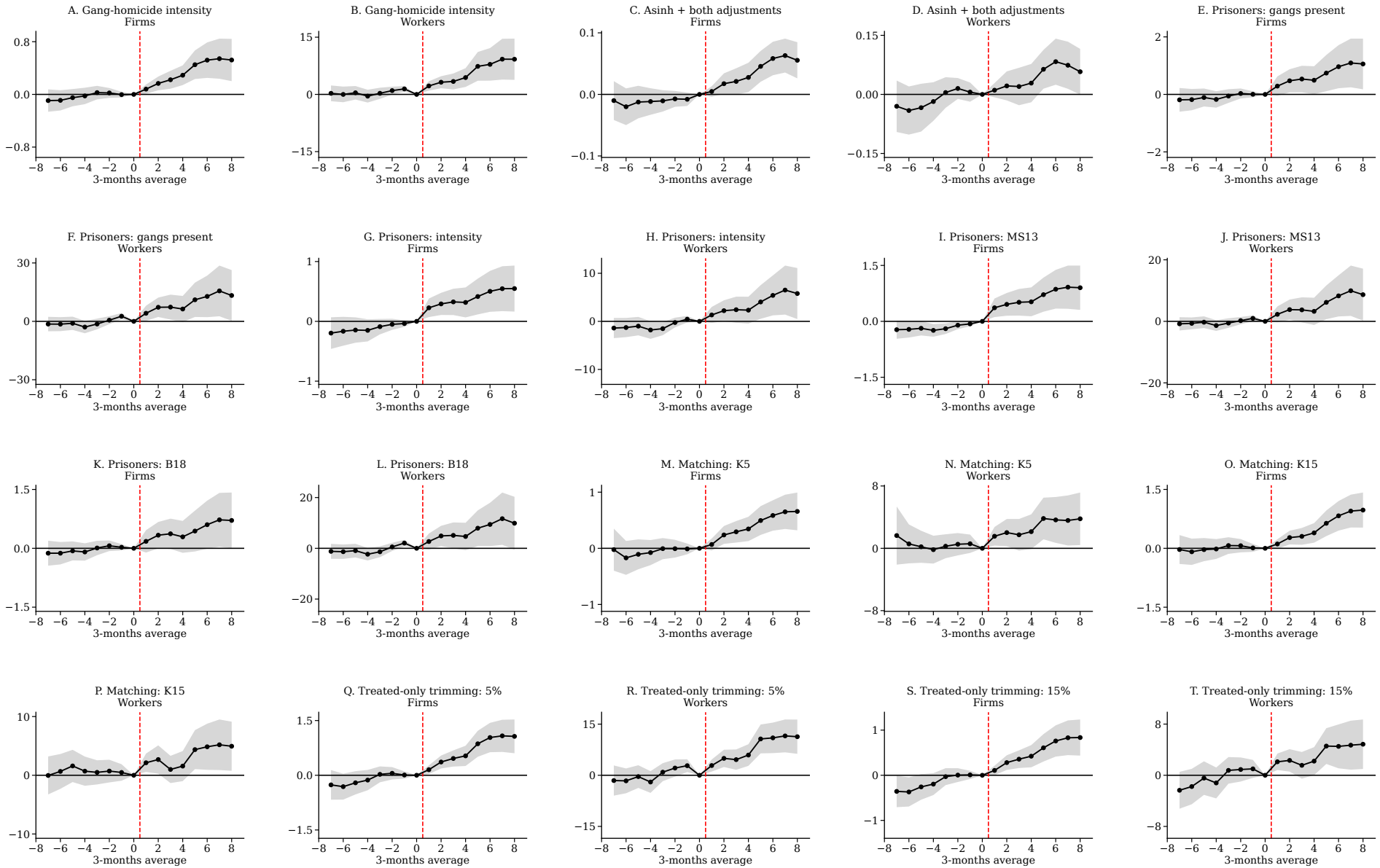
Notes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The main exposure indicator equals one for polygons with at least one recorded gang-related homicide during the pre-truce reference period. Benchmark, homicide-intensity, and panel B regressions include polygon and quarter fixed effects. Panel B separately adds firm-count-decile-by-period effects, clusters standard errors by municipality, or combines both adjustments; otherwise standard errors are clustered by polygon. Changing clustering alone leaves coefficients unchanged. Panel C uses BJS-style imputation, as described in Online Appendix Section B.1.1. Panel D uses polygon and quarter fixed effects and polygon-clustered SEs; matched samples retain control-reuse weights. NN denotes neighbors; calipers are in standard deviations of pre-truce firm counts; reuse is maximum additional uses. Treated-only trimming excludes exposed polygons above the 95th or 85th percentile of average total firm counts over the eight pre-truce periods, calculated within the exposed group.

For level outcomes, % mean is $100\hat{\beta}/\bar{Y}_{\text{pre}}$ for the exposed group (the corresponding binary-exposed group for intensity). [a]Asinh rows instead report $100\{\sinh[\text{asinh}(\bar{Y}_{\text{pre}})+\hat{\beta}]-\bar{Y}_{\text{pre}}\}/\bar{Y}_{\text{pre}}$, a back-transformation at the mean, not an average percentage treatment effect. [b]Intensity coefficients are per standard deviation of transformed exposure; outcomes remain in levels.

Gang-homicide intensity is asinh of gang-related homicides per 1,000 residents during the pre-truce reference period, standardized across polygons. Prisoner intensity analogously uses unique gang-affiliated prisoners per 1,000 residents, located by pre-incarceration residence for prison entries during the pre-truce reference period; asinh is standardized before zero-anchoring. MS13 and B18 are separate indicators.

Figure A.2: Event studies across alternative specifications



Notes: Read panels left to right, then top to bottom, with firms followed by workers for each specification. Lines connect quarterly estimates; shading shows pointwise 95% confidence intervals. $t = 0$ is normalized with no confidence interval and $t = 1$ is the first truce quarter. Axes are symmetric, with panel-specific limits. A–D use TWFE. C–D show only the asinh specification with both pre-truce firm-stock-decile-by-quarter effects and municipality clustering. A–B use polygon clustering. E–L use a BJS-style imputation specification with initial-firm-group-by-quarter effects. E–F require prisoner residences from both MS13 and B18. M–P show K5/K15 matched-sample TWFE event studies, with the same samples, control-reuse weights, and polygon-clustered inference as Table A.1, panel D. Q–T use 5% and 15% treated-only trimming, retaining all controls (TWFE; polygon-clustered SEs). Other outcomes remain in levels; intensity coefficients are per standard deviation of exposure.

Table A.2: Continuous Difference-in-Differences Estimates Based on Homicide Reductions

Dependent	**Model**	**One Year**		**Two Years**	
		Coef/SE	Per. Change	Coef/SE	Per. Change
Firms	Linear	1.01***	(4.55%)	1.93***	(8.68%)
		(0.26)		(0.47)	
	Quadratic	1.03**	(4.61%)	1.95**	(8.78%)
		(0.46)		(0.81)	
Workers	Linear	12.72***	(8.40%)	26.11***	(17.24%)
		(3.71)		(7.16)	
	Quadratic	12.90**	(8.52%)	26.46**	(17.48%)
		(6.49)		(12.50)	

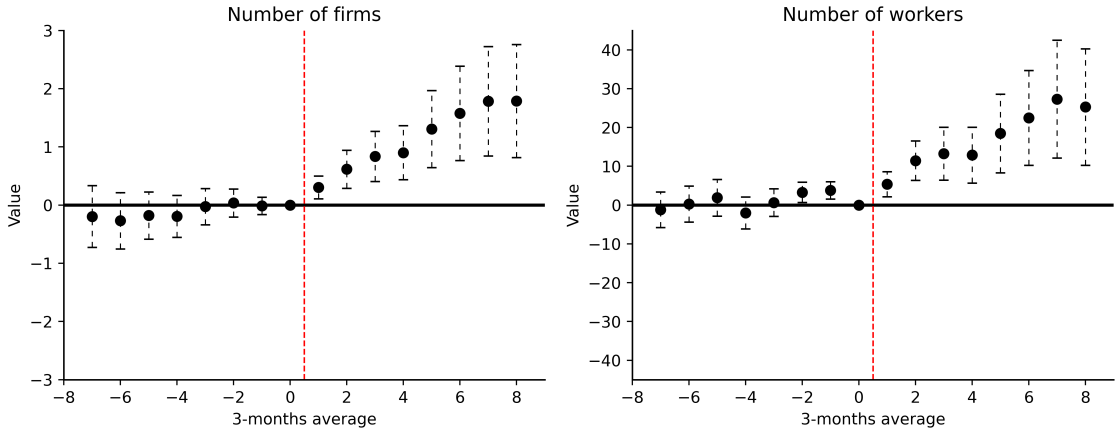
Notes: This table reports estimates from the complementary continuous specification based on the magnitude of homicide reductions during the truce. The table reports linear and quadratic specifications, with magnitudes summarized one and two years after the onset of the truce. Percentage changes are computed relative to the relevant pre-truce mean. Standard errors are shown in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.3: Predictors of Exposure to Homicide Reductions During the Truce

	Exposure to Homicide Reduction					
	(1)	(2)	(3)	(4)	(5)	(6)
Log Avg. Firms (Pre)	0.095*	0.050	—	—	0.050	0.017
	(0.055)	(0.054)			(0.055)	(0.057)
Log Avg. Workers (Pre)	0.038	0.008	—	—	0.008	-0.022
	(0.038)	(0.036)			(0.036)	(0.038)
Average Hourly Wage (Pre)	-0.086**	-0.070*	—	—	-0.067	-0.045
	(0.044)	(0.040)			(0.042)	(0.045)
Firm Growth (Pre)	—	—	0.490	1.231	1.913	-1.402
			(3.271)	(2.972)	(3.047)	(3.189)
Worker Growth (Pre)	—	—	1.573	1.091	0.557	1.631
			(1.286)	(1.183)	(1.220)	(1.333)
Wage Growth (Pre)	—	—	-1.222	0.205	-0.728	-1.584
			(2.254)	(2.127)	(2.216)	(2.707)
Log Population	—	0.269***	—	0.309***	0.269***	0.017
		(0.040)		(0.036)	(0.040)	(0.046)
Average Homicide (Pre)	—	0.036***	—	0.037***	0.036***	0.015***
		(0.004)		(0.004)	(0.004)	(0.004)
Wage Missing	-0.094	0.022	-0.245***	0.068	0.029	-0.133
	(0.107)	(0.103)	(0.067)	(0.074)	(0.104)	(0.104)
R-squared	0.035	0.122	0.011	0.118	0.122	0.464
Obs.	1581	1581	1581	1581	1581	1581

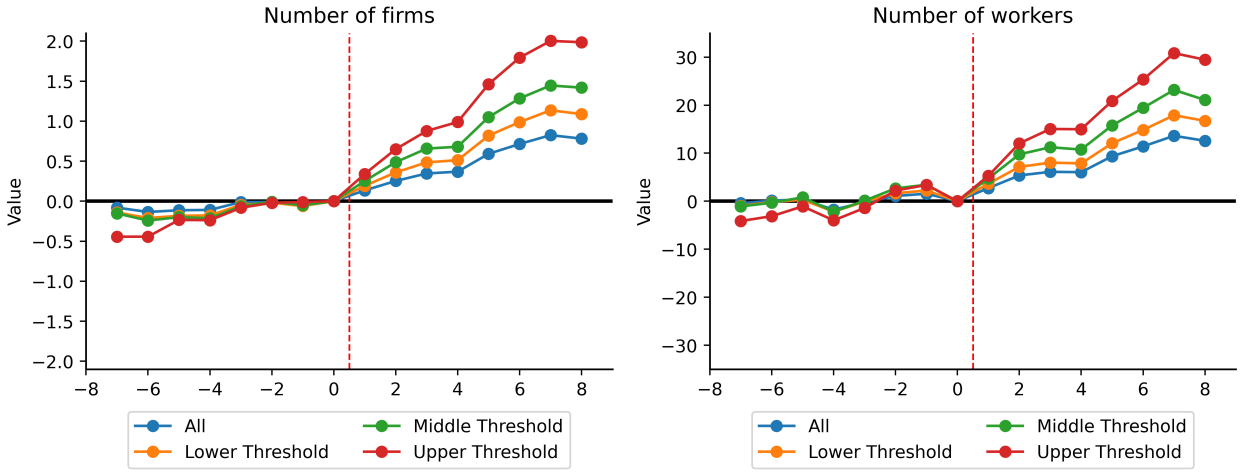
Notes: This table examines whether pre-truce economic conditions predict exposure to homicide reductions during the truce. The dependent variable is exposure to homicide reduction. Economic variables are measured before the truce. Specifications also include population and pre-truce homicide controls where indicated. Standard errors are shown in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure A.3: Event-Study Estimates for the Continuous-Treatment Specification



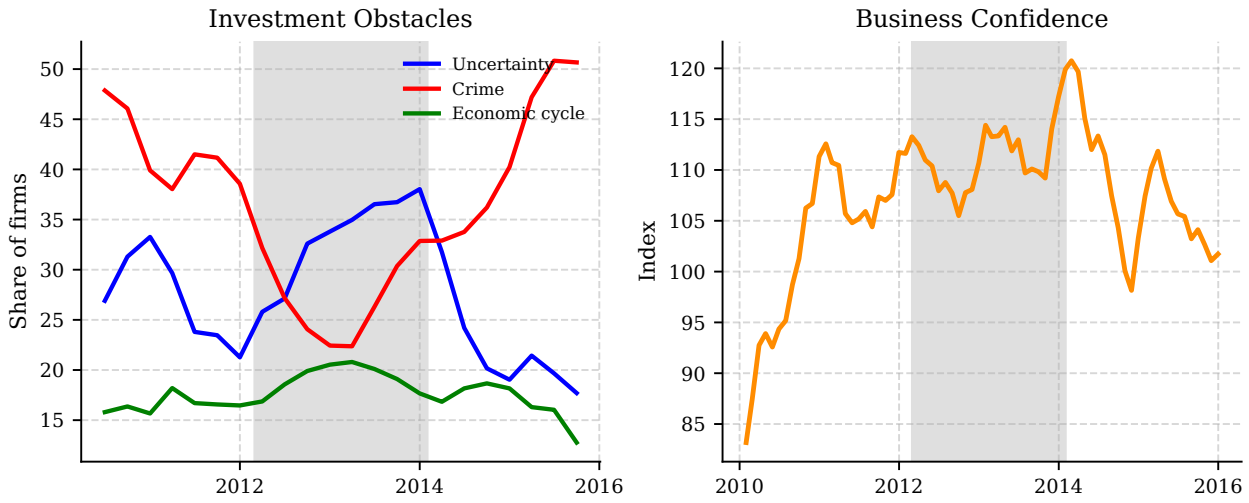
Notes: This figure presents event-study estimates under the complementary continuous-treatment specification, following the linear specification in Equation (7)*. Treatment intensity is defined as the standardized reduction in homicides during the truce period. The model includes unit and time fixed effects. Vertical bars denote 95% confidence intervals with standard errors clustered at the geographic unit level.*

Figure A.4: Event-Study Estimates by Treatment Intensity, Excluding the Top 5% Largest Firms



Notes: This figure presents estimated coefficients from Equation (1)*, applied separately to groups of polygons classified by the intensity of total homicide reductions during the truce period. The three lines correspond to Low, Intermediate, and High Threshold groups, defined respectively as polygons at or above the 25th, 50th, and 75th percentiles of the homicide-reduction distribution among treated polygons. The X-axis consolidates time into three-month intervals, spanning 24 months before and after the onset of the truce.*

Figure A.5: Investment Obstacles and Business Confidence During the Truce



Notes: The left panel plots the shares of firms citing crime, uncertainty, or the economic cycle as their main investment obstacle. The right panel plots the business confidence index. Both panels use FUSADES Economic Outlook Reports (FUSADES, 2010–2020). The shaded area indicates the truce period. These series are used as aggregate contextual evidence and are not part of the main polygon-level design.

Table A.4: Difference-in-Differences Estimates for Wages

Variables	**Main exposure**			**Larger vs. smaller homicide declines**		
	Bottom 95%	Top 5%	All firms	Bottom 95%	Top 5%	All firms
Treatment	1.396	4.995	3.928	6.169**	9.356	7.135*
	(3.135)	(9.621)	(3.825)	(3.014)	(9.966)	(3.873)
Mean	307.896	514.111	366.123	298.541	532.284	372.752
R-squared	0.8827	0.9714	0.9752	0.8829	0.9715	0.9753
Obs.	11676	5221	11808	11676	5221	11808
Clusters	734	335	738	734	335	738

Notes: This table reports difference-in-differences estimates for monthly wages per worker. The main exposure indicator equals one for polygons with at least one observed gang-related homicide before the truce and zero for polygons with no such observed homicide. The second set of columns compares polygons in the top half of the observed homicide-reduction distribution during the truce with those in the bottom half. The sample is restricted to polygons with more than 10 total employees. Wage outcomes are coded as missing when the corresponding worker denominator is zero, so the number of observations varies across firm-size categories. All regressions include unit and time fixed effects. Standard errors, clustered at the geographic unit, are shown in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.5: Spillover Estimates Using Measured Pre-Truce Gang Exposure

	Number of firms			Workers		
Variables	Base	Neighbor	Proximity	Base	Neighbor	Proximity
Treatment	1.175***	1.498***	1.031***	16.027***	19.770***	14.373**
	(0.366)	(0.364)	(0.373)	(5.747)	(5.716)	(5.793)
Spillover - neighbor		0.783***			9.079***	
		(0.167)			(1.776)	
Spillover - proximity			0.031***			0.353***
			(0.008)			(0.079)
Mean	38.314	38.314	38.314	267.543	267.543	267.543
R-squared	0.999	0.999	0.999	0.996	0.996	0.996
Obs.	25296	25296	25296	25296	25296	25296
Clusters	1581	1581	1581	1581	1581	1581

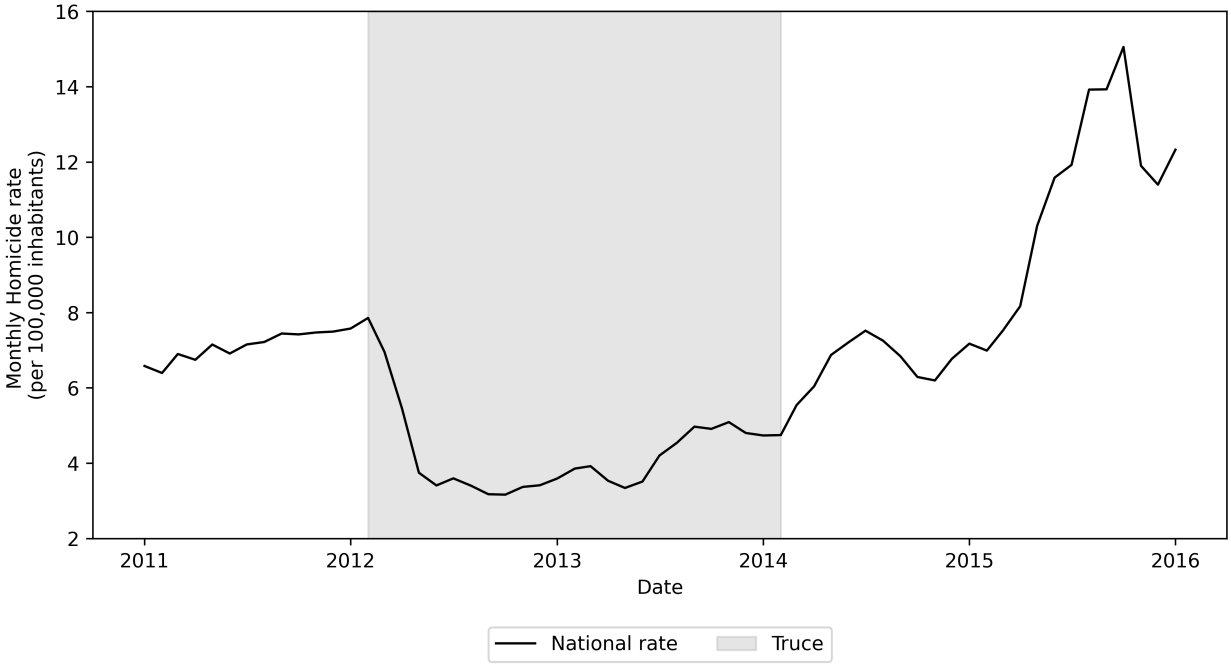
Notes: This table reports difference-in-differences estimates that include direct exposure and neighboring exposure measures. The direct exposure indicator equals one for polygons with at least one observed gang-related homicide before the truce. Spillover – neighbor equals one for polygons with spatial contact with at least one directly exposed polygon. Spillover – proximity is increasing in closeness to the nearest directly exposed polygon. All regressions include unit and time fixed effects. Standard errors are clustered at the geographic unit. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.6: Estimates Grouping Directly Exposed and Neighboring Areas

Variables	**Number of firms**	**Workers**
Treatment + Neighbor	1.041***	12.931***
	(0.174)	(2.382)
Mean	24.873	171.814
R-squared	0.999	0.996
Obs.	25296	25296
Clusters	1581	1581

Notes: This table reports difference-in-differences estimates from a specification that redefines exposure to include both polygons with measured pre-truce gang exposure and polygons with spatial contact with exposed polygons. This grouped exposure definition addresses the concern that neighboring areas may be indirectly exposed to the truce-related improvement in security. The mean is computed for the combined directly exposed and neighboring group before the truce. All regressions include unit and time fixed effects. Standard errors are clustered at the geographic unit. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure A.6: National Homicide Rate Around and After the Truce



Notes: This figure plots the national monthly homicide rate per 100,000 inhabitants. The series is smoothed using a trailing 3-month moving average. The shaded area highlights the duration of the truce.

B Online Appendix

B.1 Additional Specifications

B.1.1 Imputation-Based Estimation

To estimate responses using the prisoner-based exposure definitions in Appendix Table A.1, panel C, I refine the specification by allowing for group-specific time effects based on the initial number of firms. Specifically, I divide polygons into two groups according to the number of firms in the first pre-truce period and interact group membership with each three-month period. This approach allows areas with larger and smaller initial business bases to follow different outcome paths. The exact grouping rule is documented in the replication materials.

For each binary exposure definition, let D_i indicate measured exposure and define $D_{it} = D_i \times \mathbf{1}\{t \geq t_0\}$, where t_0 denotes the onset of the truce. The counterfactual outcome model is

$$Y_{it}(0) = \gamma_i + \lambda_t + \omega_{g(i),t} + \varepsilon_{it}. \tag{2}$$

In this equation, $Y_{it}(0)$ denotes the counterfactual outcome for polygon i at time t. The terms γ_i and λ_t represent polygon and period fixed effects, respectively, while $\omega_{g(i),t}$ captures group-by-period effects, normalized to zero for the lower-initial-firm group. Unlike a group-specific linear trend, these interactions allow the difference between the two groups' time paths to vary freely across periods.

Following Borusyak, Jaravel and Spiess (2024), I estimate the fixed effects and group-by-period effects using observations with $D_{it} = 0$: all pre-truce observations, including exposed polygons before the truce, and all observations from polygons without the corresponding measured exposure. I then use these estimates to impute counterfactual outcomes for exposed observations in the post-truce period. The estimated response for each exposed observation is the difference between its observed and imputed counterfactual outcome:

$$\widehat{\tau}_{it} = Y_{it} - \widehat{\gamma}_i - \widehat{\lambda}_t - \widehat{\omega}_{g(i),t}. \tag{3}$$

For binary exposure, the period-specific estimate averages these residuals across exposed polygons. The table reports the equally weighted average over the eight post-truce periods. As in the main

specification, these estimates describe differential responses by measured exposure rather than the total effect of the truce relative to unaffected locations.

For the continuous prisoner-exposure specification, the first-stage counterfactual model uses all pre-truce observations and all observations from polygons with zero measured exposure. Let $d_i \geq 0$ denote the exposure dose, anchored at zero for polygons without measured exposure and expressed in standard-deviation units. In each post-truce period, I project the imputed residuals on the dose without an intercept among exposed polygons:

$$\widehat{\beta}_t = \frac{\sum_{i:d_i>0} d_i \widehat{\tau}_{it}}{\sum_{i:d_i>0} d_i^2}. \tag{4}$$

The table reports the equally weighted average of $\widehat{\beta}_t$ over the eight post-truce periods. This coefficient summarizes the relationship between imputed outcome differences and exposure intensity; it is not a binary-exposure average treatment effect.

Standard errors are determined using a polygon-level block bootstrap with 999 replications. Each replication resamples entire polygon histories and re-estimates the counterfactual model, holding the exposure definitions and initial-firm groups fixed. Period-specific standard errors are the standard deviations of the corresponding bootstrap estimates; the standard error of the average response is computed from the averages within each replication. Confidence intervals and two-sided tests use a normal approximation.

The pre-truce coefficients displayed in Appendix Figure A.2, panels E–L, are estimated separately using a saturated exposure-by-period regression with the same group-by-period controls and polygon-clustered standard errors. They provide pre-trend diagnostics rather than imputed treatment effects. The final pre-truce period is the omitted reference period, normalized to zero; post-truce coefficients are the imputation estimates described above.

B.1.2 Continuous Treatment Specification

This subsection provides additional details on the continuous-treatment specification used as a complement to the baseline binary exposure design in the main text. While the main empirical strategy relies on a proxy for pre-truce gang exposure, this complementary exercise examines whether areas experiencing larger realized declines in violence during the truce also exhibit stronger economic responses.

A key concern in this setting is that, when treatment intensity varies continuously, standard two-

way fixed effects (TWFE) estimators may be difficult to interpret in the presence of heterogeneous or nonlinear effects (Callaway, Goodman-Bacon and Sant'Anna, 2024). For this reason, I implement a residualization approach that separates the variation used for identification from fixed differences across polygons and common aggregate shocks over time.

Let $D_i \in \mathbb{R}_+ \cup \{0\}$ denote treatment intensity, where $D_i = 0$ for polygons with no realized homicide decline relative to the reference cutoff, and $D_i > 0$ indexes the magnitude of the observed decline during the truce. In a first step, I estimate the following model using untreated observations:

$$Y_{it} = \alpha_i + \lambda_t + \epsilon_{it} \tag{5}$$

where Y_{it} denotes the outcome of interest for polygon i at time t, α_i are polygon fixed effects, and λ_t are time fixed effects. I then construct residualized outcomes by removing these estimated fixed effects from the observed data:

$$\tilde{Y}_{it} = Y_{it} - \hat{\alpha}_i - \hat{\lambda}_t \tag{6}$$

In a second step, I relate these residualized outcomes to treatment intensity using a flexible dose-response specification. In the empirical implementation, this relationship is estimated using linear and quadratic specifications of treatment intensity, allowing the response of outcomes to vary with the magnitude of the homicide decline.

Let $\Delta y_t(d)$ denote the residualized outcome response at event time t for treatment intensity d. I estimate two parametric dose-response functions. The linear specification is

$$\Delta y_t(d) = \beta_1 d, \tag{7}$$

while the quadratic specification is

$$\Delta y_t(d) = \beta_1 d + \beta_2 d^2. \tag{8}$$

In the linear specification, the marginal response to treatment intensity is constant and given by β_1. In the quadratic specification, the marginal response varies with intensity and is given by $\beta_1 + 2\beta_2 d$.

This continuous-treatment analysis is interpreted as complementary evidence rather than as the paper's primary identification strategy. It is useful for characterizing how economic outcomes covary

with larger observed reductions in violence, but it relies on stronger assumptions than the baseline binary specification and is therefore interpreted more cautiously.

B.2 Construction of the Homicide Reduction Index

To systematically classify treatment and control areas, I construct a standardized measure of abnormal homicide declines. Let $H_{p,t}$ denote the homicide rate in polygon p at month t. I first compute a short-run change in the homicide rate as follows:

$$\Delta H_{p,t} = \left(\frac{1}{36} \sum_{\tau=t-36}^{t-1} H_{p,\tau} \right) - \left(\frac{1}{3} \sum_{\tau=t}^{t+2} H_{p,\tau} \right).$$

This measure $\Delta H_{p,t}$ compares the mean homicide rate over the 36 months prior to t with the mean rate after the event window. In the empirical implementation, I omit the immediate transition months around the onset of the truce before measuring the post-event decline. During the truce, I expect some polygons to exhibit larger values of $\Delta H_{p,t}$, indicating a sharp reduction in total homicide rates.

Next, I standardize each observed drop against the distribution of comparable homicide changes within the same polygon. Define the polygon-specific mean and standard deviation of these changes:

$$\mu_p = \mathbb{E}[\Delta H_{p,t'}], \quad \sigma_p = \sqrt{\mathbb{E}[(\Delta H_{p,t'} - \mu_p)^2]},$$

Using these statistics, I construct a standardized measure:

$$Z_{p,t} = \frac{\Delta H_{p,t} - \mu_p}{\sigma_p}.$$

This $Z_{p,t}$ score indicates how unusual the homicide reduction is for polygon p at time t, relative to its own historical variability. A more positive $Z_{p,t}$ means that the current drop in homicides is large relative to what would typically be expected in that polygon based on past fluctuations.

B.2.1 Placebo tests

I apply the homicide-decline classification at the actual truce date and at placebo dates 3, 5, 7, and 9 months earlier. These datasets are stacked by assignment scenario s. Let R_s indicate the actual-truce scenario. The specification includes polygon-by-scenario fixed effects and event-time fixed effects that differ between the actual-truce scenario and the pooled placebo scenarios.

The stacked TWFE specification interacts treatment with an indicator for the actual-truce assignment scenario:

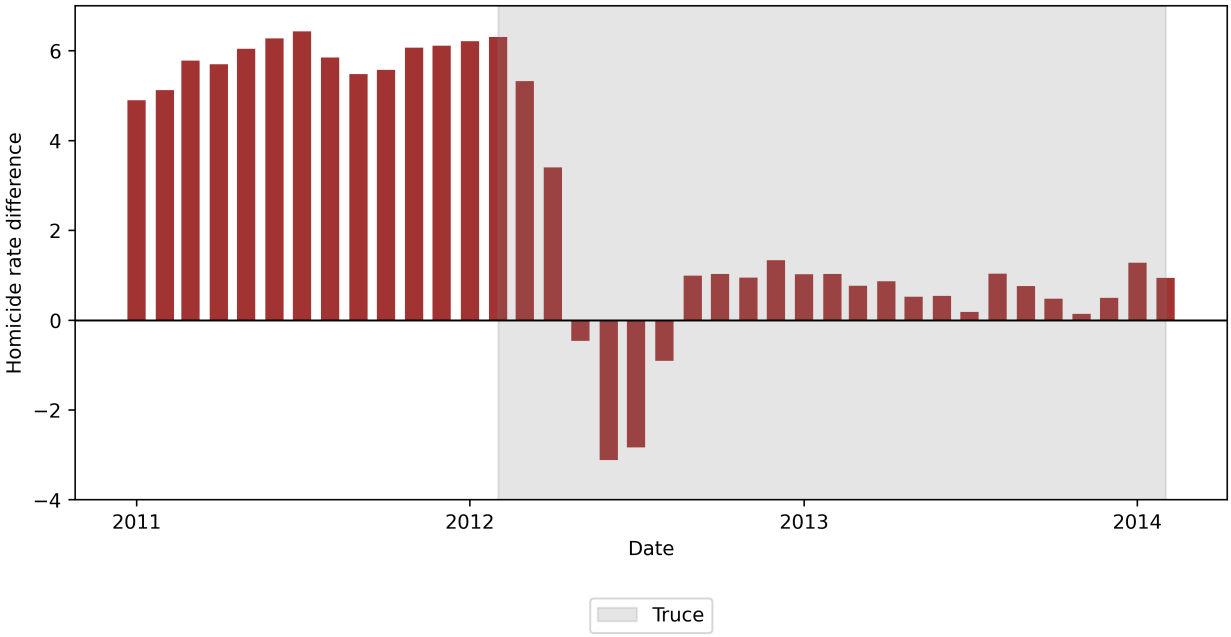
$$Y_{ist} = \alpha + \tau_0 D_{ist} + \tau_1 D_{ist} R_s + \gamma_{is} + \lambda_t + \theta_t R_s + \epsilon_{ist} \tag{9}$$

where Y_{ist} represents the outcome of interest for polygon i in assignment scenario s at event time t, measured relative to the assignment date and covering periods before and after it. R_s equals one for the actual-truce scenario and zero for placebo scenarios. D_{ist} equals one for treated polygons after the assignment date in scenario s. The terms γ_{is} are polygon-by-scenario fixed effects, while $\lambda_t + \theta_t R_s$ allows event-time fixed effects to differ between the actual-truce scenario and the pooled placebo scenarios. The term ϵ_{ist} is the idiosyncratic error term. The errors are clustered at the level of geographic units interacted with s. The coefficient τ_0 captures the pooled difference-in-differences estimate across scenarios with assignment dates preceding the actual truce. The coefficient τ_1 captures the incremental difference-in-differences estimate for the actual-truce scenario relative to the pooled placebo scenarios.

I also estimate an event-study specification by interacting exposure with event-time indicators.

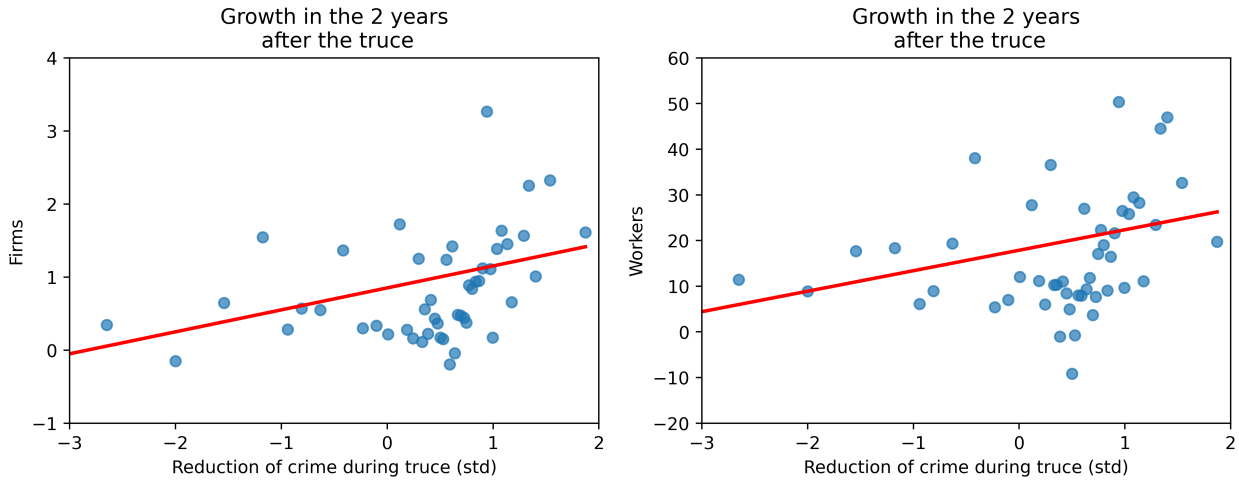
B.3 Figures and Tables

Figure B.1: Difference in Homicide Rates by Realized Homicide Reductions



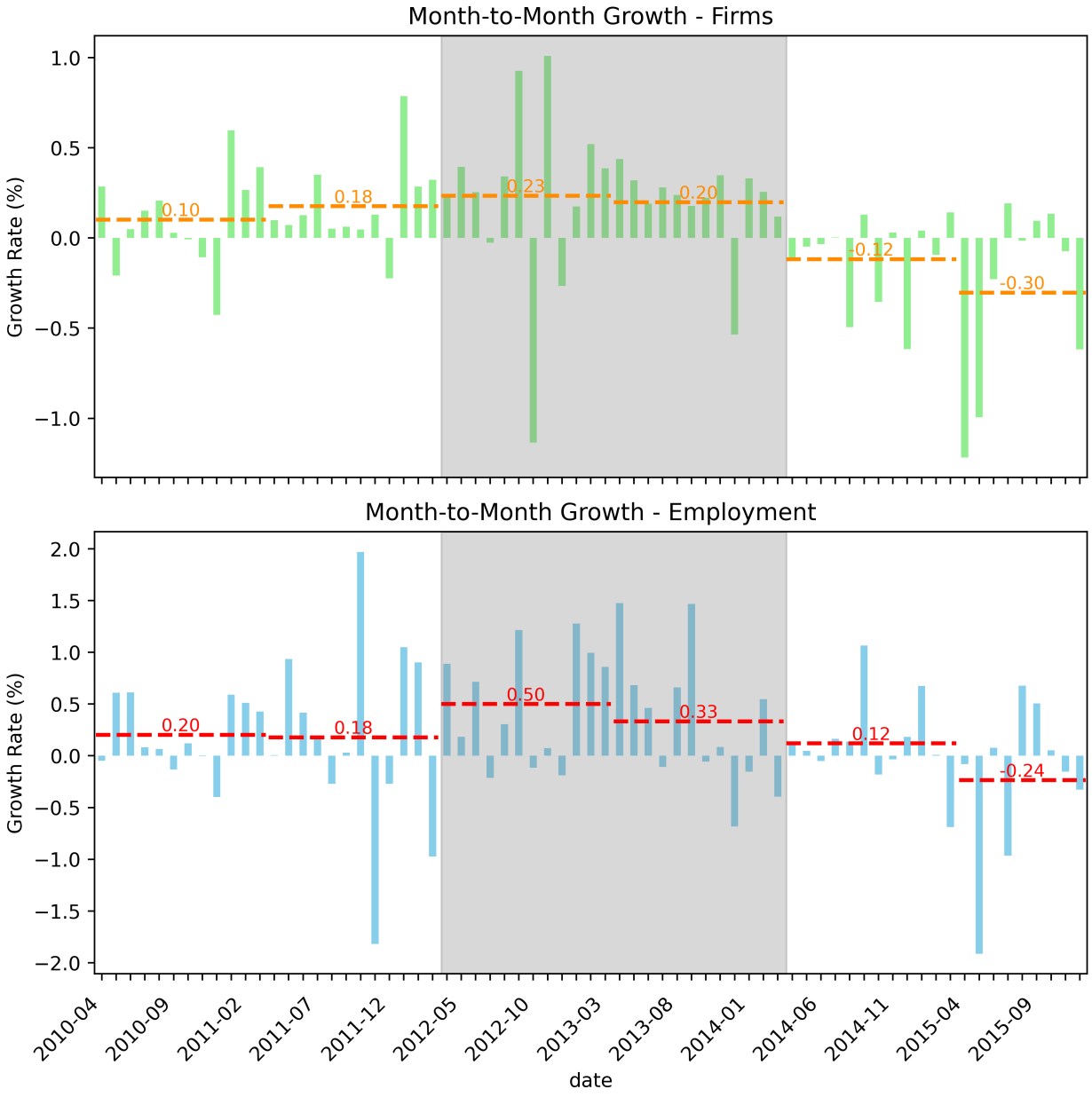
Note: This figure plots the monthly difference in homicide rates between areas classified as treated and control based on realized homicide reductions during the truce. The underlying homicide-rate series are smoothed using a trailing 3-month moving average. The bars summarize how the homicide-rate gap evolves around the onset of the truce.

Figure B.2: Relationship Between Crime Reduction During the Truce and Economic Activity



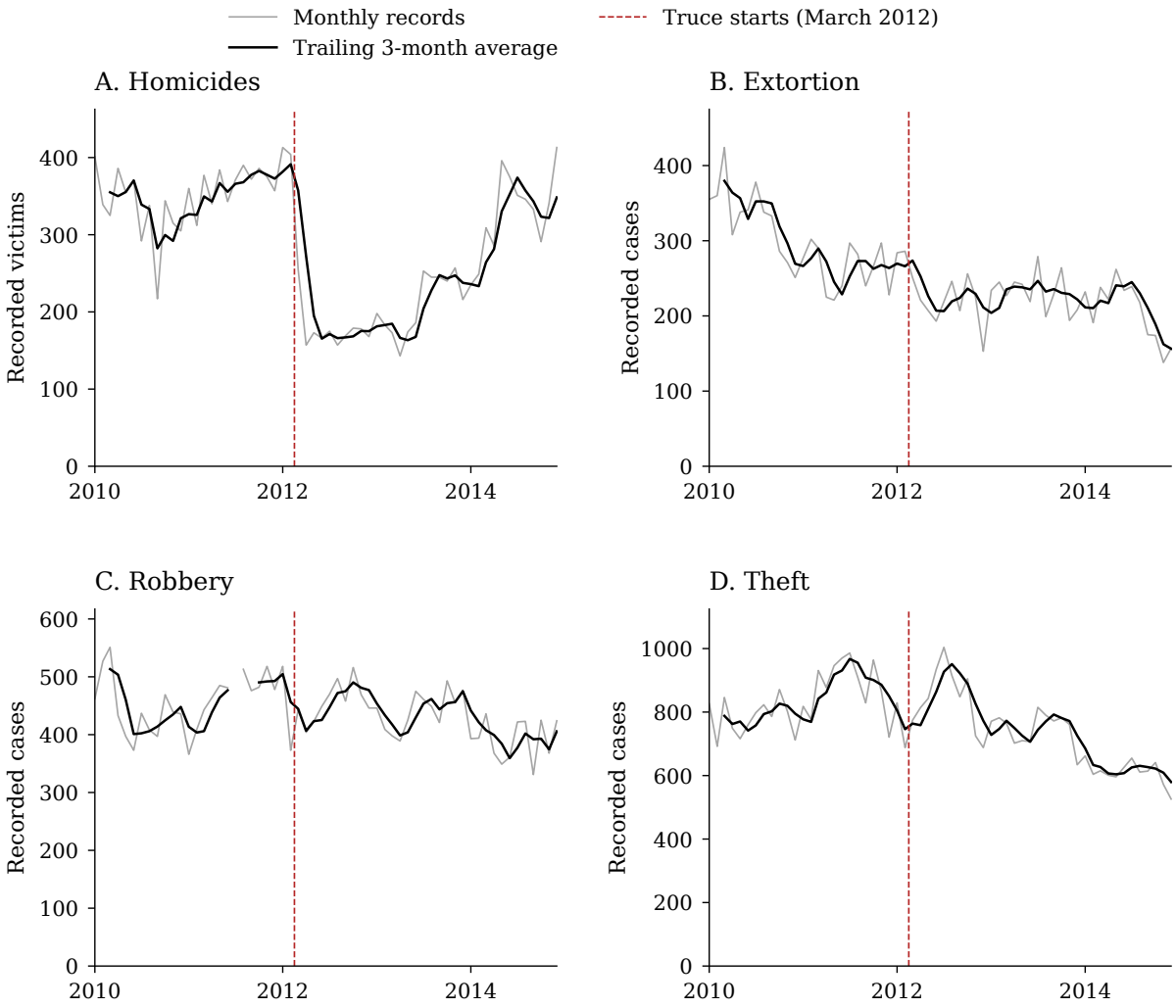
Notes: This figure presents the relationship between the reduction in crime during the truce and two key economic indicators: firm presence (left panel) and employment (right panel). Each point represents a bin of 50 observations, with a fitted linear regression line shown in red. The left and right panels show changes in firm counts and employment, respectively, against observed homicide reductions.

Figure B.3: Monthly Growth in Firms and Employment



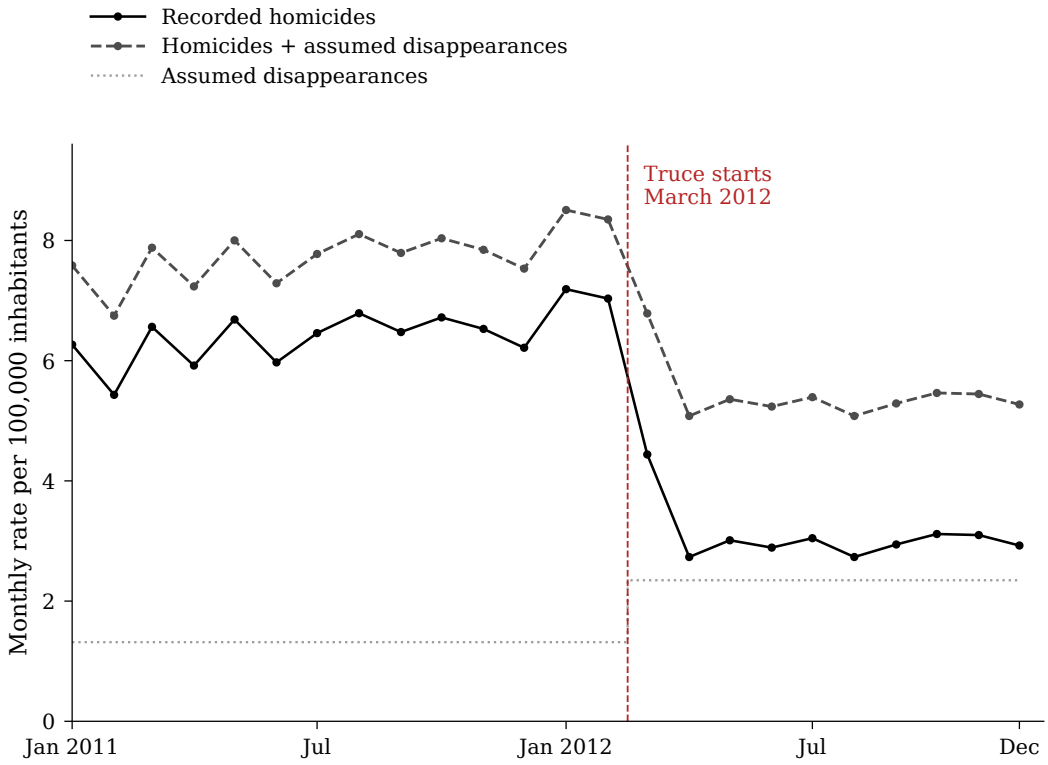
Notes: This figure presents the monthly percentage growth of firms (top panel) and employment (bottom panel) over the study period. Each bar represents the month-to-month growth rate, while the dashed red and orange lines indicate the average growth rates for specific periods. The shaded gray area highlights the truce period.

Figure B.4: National Crime Trends Around the Truce



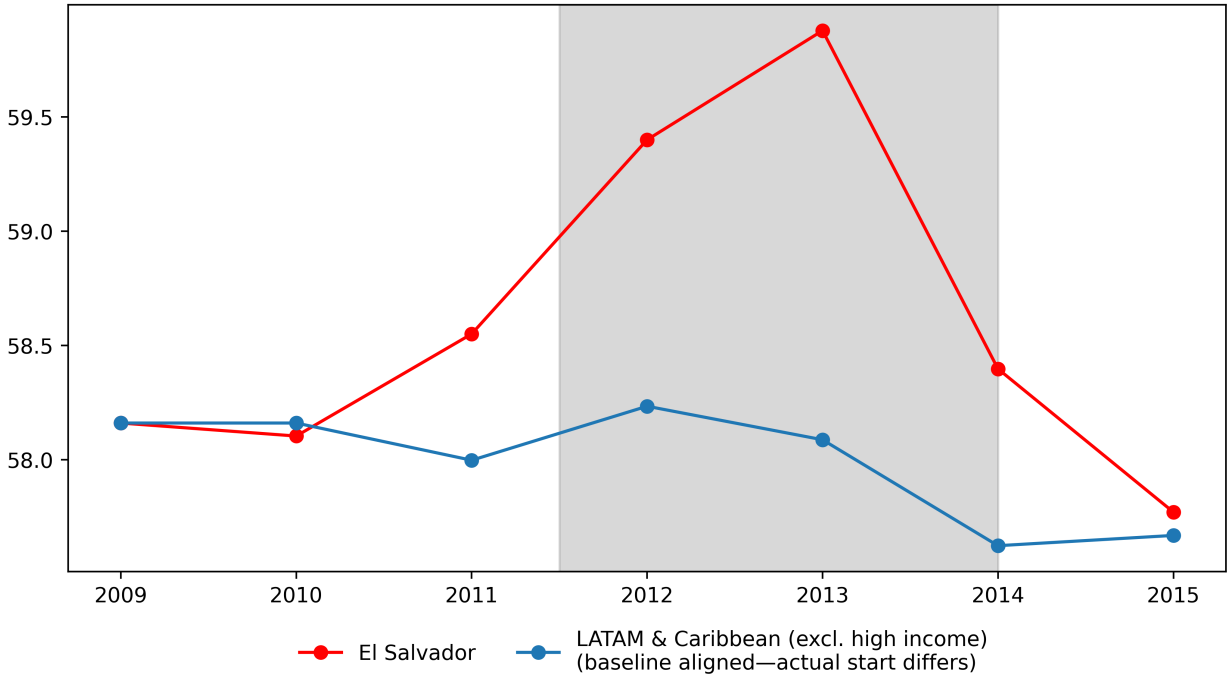
Notes: The panels plot national monthly records of homicides, extortion, robbery, and theft from January 2010 to December 2014. Gray lines show monthly records; black lines show trailing three-month averages. Red dashed lines mark the transition between February and March 2012, when the truce began. Sources are the police records and the national homicide records. July 2011 robbery is omitted because of incomplete dates in the source records. These are recorded offenses, which may differ from actual incidence.

Figure B.5: Homicides Under an Assumed Increase in Disappearances at Truce Onset



Notes: This figure considers a hypothetical scenario in which all recorded disappearances are additional homicides, with no overlap with recorded homicide victims. Disappearance data from Carcach and Artola (2016) are available only annually. I distribute the 2011 rate uniformly across months and maintain that monthly level through February 2012. I assign the entire increase between 2011 and 2012 uniformly to March–December 2012, creating a level increase at the start of the truce. This is a strong timing assumption: recorded disappearances had already increased before the truce, and the annual data do not establish when the additional increase in 2012 occurred. The adjusted series adds this assumed monthly rate to recorded homicide rates, calculated using the national population from the 2007 Census, held fixed throughout the period. Even under these strong assumptions, the adjusted series shows a clear decline at the start of the truce. This is a sensitivity scenario based on recorded annual disappearances.

Figure B.6: Employment to Population Ratio (%) Over Time Using WDI Data



Notes: This figure presents a key labor market indicator for the period 2009–2015, using data from the World Development Indicators (WDI), which are based on household surveys. The graph displays the employment-to-population ratio (15+, total %), illustrating the proportion of the working-age population engaged in employment each year. The shaded area corresponds to the period of the gang truce (2012–2014). For comparison, the figure also includes the regional trend for Latin America and the Caribbean (excluding high-income countries), baseline-aligned to El Salvador's truce period.

Table B.1: Employment-to-Population Ratios: El Salvador and Regional Comparison

	El Salvador	Latin America and the Caribbean	Difference (percentage points)
Panel A. Annual employment-to-population ratios (%)			
2009	58.16	60.16	-2.00
2010	58.10	60.16	-2.06
2011	58.55	60.00	-1.45
2012	59.40	60.24	-0.84
2013	59.88	60.09	-0.21
2014	58.40	59.63	-1.23
2015	57.77	59.67	-1.90
Panel B. Period averages and changes			
Pre-truce: 2009–2011 (%)	58.27	60.11	-1.84
Truce: 2012–2014 (%)	59.22	59.99	-0.76
Change (percentage points)	0.95	-0.12	1.08

Notes: Data are from the World Development Indicators (WDI), based on household surveys. The employment-to-population ratio is the percentage of the population aged 15 and older that is employed. The regional aggregate excludes high-income countries. Panel A reports annual ratios; Panel B reports averages for 2009–2011 (pre-truce) and 2012–2014 (truce). The difference column is El Salvador minus the regional aggregate. The change row is the truce-period average minus the pre-truce average, in percentage points.

Table B.2: Summary statistics: main, matched, and trimmed samples

	Main sample		$NN(5)$		$NN(15)$		Trimming 5%		Trimming 15%	
	Measured pre-truce gang exposure									
	Lower	Higher	Lower	Higher	Lower	Higher	Lower	Higher	Lower	Higher
Panel A. Pre-truce economic activity										
Registered firms	9.81	38.59	8.65	11.57	12.08	13.13	9.81	23.62	9.81	14.87
	(29.39)	(122.91)	(16.43)	(15.66)	(16.62)	(16.31)	(29.39)	(30.41)	(29.39)	(15.46)
Bottom 95%	9.28	36.32	8.28	11.12	11.60	12.51	9.28	22.45	9.28	14.21
	(26.96)	(113.97)	(15.59)	(15.14)	(15.93)	(15.54)	(26.96)	(28.73)	(26.96)	(14.71)
Formal workers	168.41	735.71	128.99	158.60	165.99	212.07	168.41	398.16	168.41	232.48
	(736.72)	(2,773.39)	(391.15)	(324.87)	(359.65)	(461.07)	(736.72)	(833.97)	(736.72)	(478.50)
Bottom 95%	63.95	267.91	53.31	71.59	73.92	83.11	63.95	152.12	63.95	91.20
	(219.33)	(942.32)	(111.16)	(111.08)	(111.58)	(114.45)	(219.33)	(216.66)	(219.33)	(106.20)
Average wage	321.72	356.73	314.27	312.69	328.54	337.28	321.72	347.63	321.72	329.66
	(227.38)	(211.91)	(192.49)	(149.48)	(205.93)	(221.69)	(227.38)	(212.59)	(227.38)	(172.17)
Bottom 95%	276.58	302.12	274.18	281.34	278.54	288.39	276.58	295.80	276.58	286.79
	(103.31)	(98.15)	(102.04)	(90.20)	(92.16)	(95.93)	(103.31)	(94.80)	(103.31)	(93.59)
Firm openings per quarter	0.30	0.99	0.26	0.34	0.37	0.40	0.30	0.64	0.30	0.42
	(1.22)	(3.72)	(0.83)	(0.73)	(0.97)	(0.86)	(1.22)	(1.15)	(1.22)	(0.79)
Firm closures per quarter	0.18	0.79	0.16	0.27	0.21	0.26	0.18	0.53	0.18	0.34
	(0.68)	(2.46)	(0.55)	(0.61)	(0.56)	(0.52)	(0.68)	(0.95)	(0.68)	(0.65)
Panel B. Differential decline in homicides during the truce										
Higher minus lower exposure	0.97		0.85		0.68		0.98		1.04	
Annualized (per 100,000)	11.69		10.16		8.18		11.75		12.45	
As % of the initial homicide gap	129.5%		135.6%		102.9%		130.4%		115.3%	
Panel C. Sample size and design										
Distinct polygons	1,283	298	1,065	213	810	225	1,283	283	1,283	253
Caliper (SD)	—		0.35		0.25		—		—	
Maximum additional control reuse	—		0		4		—		—	

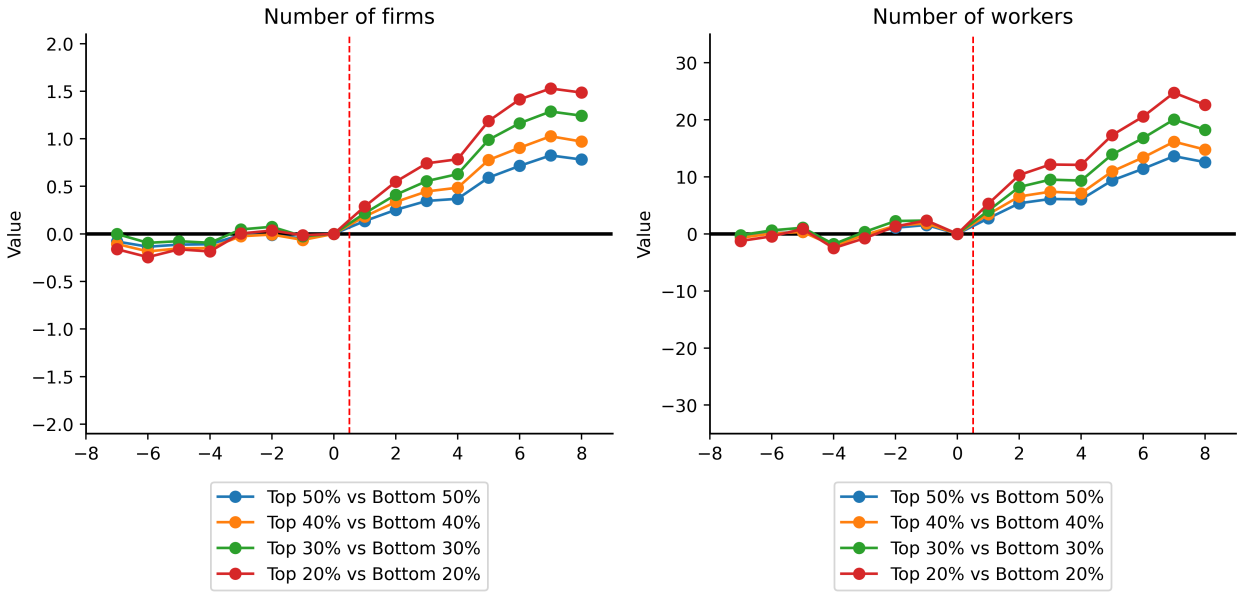
Notes: Panel A reports means for the final pre-truce three-month period, with standard deviations in parentheses below. Lower and higher refer to measured pre-truce gang exposure under the main binary definition. Within each matched sample, the lower-exposure column contains matched controls and the higher-exposure column contains the retained exposed polygons. Bottom 95% excludes the largest 5% of firms; firm openings refer to entries in the formal-firm registry. Panel B reports $(H_{\text{pre}} - H_{\text{truce}})_{\text{higher}} - (H_{\text{pre}} - H_{\text{truce}})_{\text{lower}}$, where H is the mean monthly homicide rate per 100,000 residents. Positive values indicate a larger decline in higher-exposure areas. The annualized difference equals 12 times the monthly contrast, rather than a rate for a specific calendar year. Matched-control statistics retain the original matching weights; polygon counts refer to distinct locations. $NN(k)$ denotes k-nearest-neighbor matching on pre-truce firm counts. Calipers are in standard deviations of pre-truce firm counts; maximum additional reuse is the permitted number of uses beyond the first match. Trimming removes exposed polygons above the 95th or 85th percentile of average total firm counts over the eight pre-truce periods, calculated within the exposed group. Selection is fixed over time; trimmed samples are unweighted and retain the original firm-level bottom-95% classification. Means and SDs use nonmissing observations.

Table B.3: Matching Balance Summary Across Matched Specifications ($NN(5)$ and $NN(15)$)

Spec.	Treated mean	Matched control mean	Standardized mean diff.	Control polygons used	Max reuse
K5, c=0.35, r=0	11.479	8.560	0.183	1065	0
K15, c=0.25, r=4	13.084	11.956	0.069	810	4

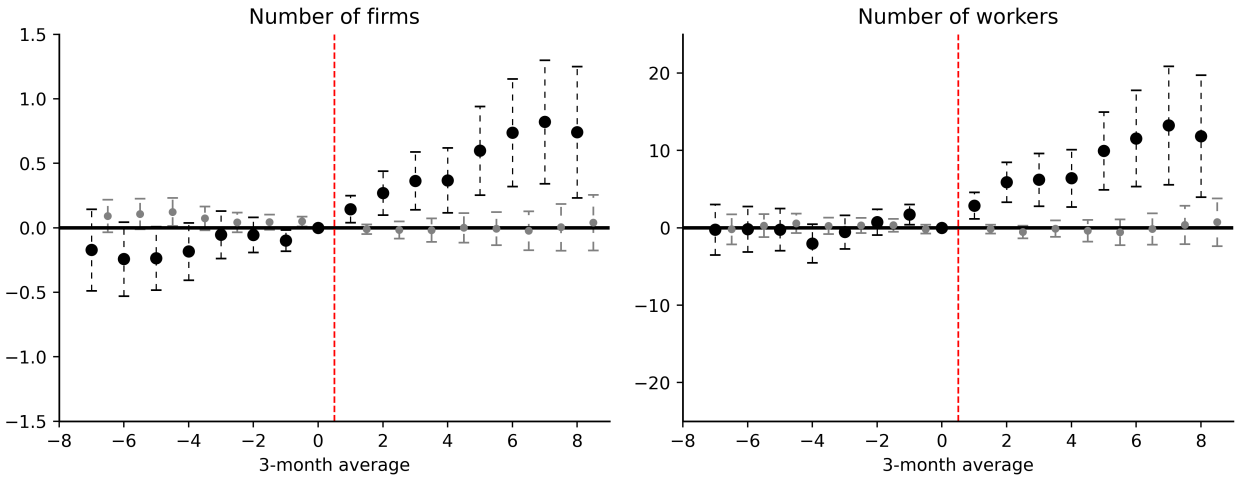
Notes: This table summarizes covariate balance across the matched specifications reported in this appendix. $NN(k)$ denotes k-nearest-neighbor matching on pre-truce firm counts. The table compares the treated-group mean with the corresponding matched-control means and reports standardized differences after matching for the $NN(5)$ and $NN(15)$ designs. Lower standardized differences indicate closer balance between treated and matched-control polygons.

Figure B.7: Event-Study Results for Different Treatment-Intensity Thresholds



Notes: This figure presents event-study estimates based on Equation (1)*, comparing outcomes across treated units with varying degrees of treatment intensity. Specifically, it plots separate estimates for units in the top 50% vs. bottom 50%, top 40% vs. bottom 40%, top 30% vs. bottom 30%, and top 20% vs. bottom 20% of the homicide reduction distribution during the truce period. The X-axis aggregates data into three-month intervals spanning 24 months before and after the truce. All regressions include unit and time fixed effects.*

Figure B.8: Event-Study Results for Placebo Truce Assignments



Notes: This figure presents event-study estimates based on the specification in Equation (9)*, which compares the estimated treatment effects for the actual truce period versus placebo periods starting 3, 5, 7, and 9 months prior. Each dataset defines treatment based on abnormal homicide reductions, and is stacked with a scenario indicator* s *identifying the truce or placebo period. Polygon fixed effects are interacted with assignment scenario; event-time fixed effects differ between the actual-truce scenario and the pooled placebo scenarios. The X-axis consolidates time into three-month intervals spanning 24 months before and after the treatment assignment. Black markers report the incremental estimates for the actual-truce scenario relative to the pooled placebo scenarios; gray markers report the pooled placebo estimates. Vertical bars represent 95% confidence intervals, with standard errors clustered at the unit* $\times$ *scenario level.*

Table B.4: Sensitivity of Treatment Effects to Alternative Sample Cutoffs

	Homicide-reduction sample cutoff				
Top / bottom share	50% / 50%	40% / 40%	30% / 30%	20% / 20%	10% / 10%
Panel A. Registered firms					
Treatment	0.566***	0.727***	0.833***	1.088***	0.677
	(0.184)	(0.225)	(0.289)	(0.395)	(0.415)
% of pre-truce mean	2.544	2.854	2.785	3.083	2.564
R^2	0.9986	0.9987	0.9987	0.9989	0.9965
Panel B. Formal workers					
Treatment	8.313***	9.884***	11.939***	15.681***	9.997**
	(2.537)	(3.130)	(4.053)	(5.814)	(4.297)
% of pre-truce mean	5.490	5.666	5.778	6.380	5.898
R^2	0.9958	0.9959	0.9961	0.9962	0.9930
Observations	25,296	19,808	14,848	9,920	5,104

Notes: This table reports difference-in-differences estimates using a two-way fixed effects specification analogous to Equation (1). Each column compares units in the top and bottom indicated shares of homicide reductions during the truce. Panel A reports registered firms and Panel B formal workers; both exclude the top 5% of firms by size. All regressions include unit and time fixed effects. Standard errors, clustered by geographic unit, appear in parentheses below coefficients. Effects relative to the pre-truce baseline mean are expressed in percent. Observations are identical across outcomes within each column. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table B.5: Placebo Triple-Differences Estimates

Variables	**3 months before**	**5 months before**	**7 months before**	**9 months before**	**All placebos**
			Number of firms		
Treatment (Placebo)	0.102	-0.146	-0.129	-0.108	-0.070
	(0.170)	(0.160)	(0.155)	(0.148)	(0.079)
Treatment x Truce	0.463*	0.711***	0.694***	0.674***	0.636***
	(0.250)	(0.244)	(0.241)	(0.236)	(0.200)
R-squared	0.9987	0.9987	0.9988	0.9988	0.9988
Obs.	50592	50592	50592	50592	126480
			Workers		
Treatment (Placebo)	-0.387	-0.112	0.038	-0.641	-0.275
	(2.060)	(1.841)	(1.610)	(1.394)	(0.872)
Treatment x Truce	8.700***	8.424***	8.274***	8.953***	8.588***
	(3.267)	(3.134)	(3.004)	(2.894)	(2.682)
R-squared	0.9961	0.9964	0.9965	0.9966	0.9968
Obs.	50592	50592	50592	50592	126480

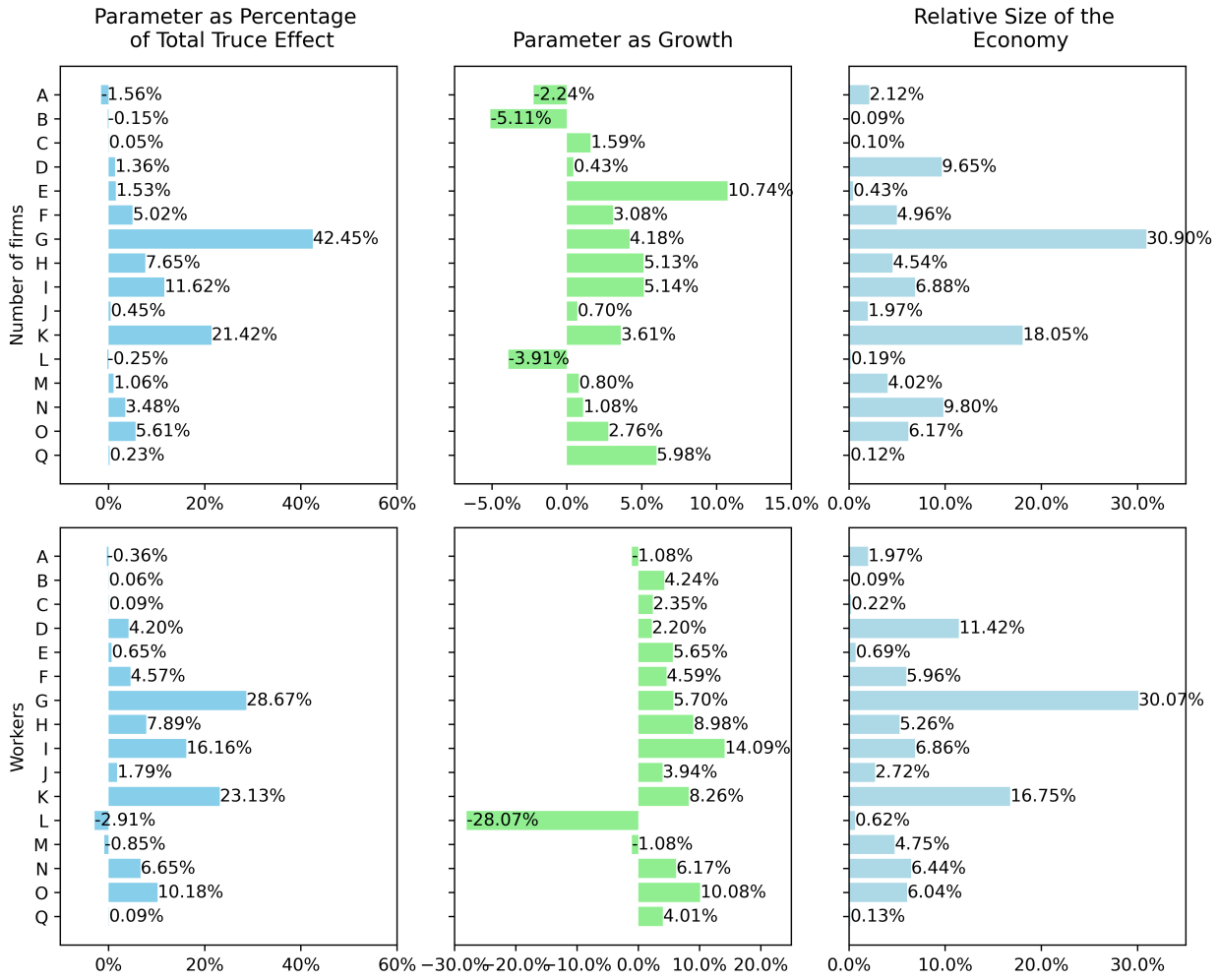
Notes: This table reports the triple-differences estimates examining the effect of the truce and several placebo periods set 3, 5, 7, and 9 months prior to the actual onset of the truce. Each column presents results from a separate specification. The rows labeled "Treatment (Placebo)" report the pooled difference-in-differences estimates for assignment dates preceding the actual truce. The row "Treatment x Truce" reports the incremental estimate for the actual-truce scenario relative to those placebo scenarios. Standard errors (in parentheses) are clustered at the polygon level interacted with placebo period or truce. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Table B.6: Placebo Triple-Differences Estimates by Treatment Intensity

Variables	Treatment	Lower Threshold	Middle Threshold	Upper Threshold
		Number of firms		
Treatment (Placebo)	-0.070	-0.011	0.104	0.164
	(0.079)	(0.085)	(0.101)	(0.124)
Treatment x Truce	0.636***	0.814***	0.921***	1.283**
	(0.200)	(0.244)	(0.333)	(0.559)
R-squared	0.9988	0.9989	0.9989	0.9989
Obs.	126480	111024	95616	80256
		Workers		
Treatment (Placebo)	-0.275	0.363	1.830	3.638***
	(0.872)	(0.952)	(1.142)	(1.389)
Treatment x Truce	8.588***	10.573***	12.239**	16.613*
	(2.682)	(3.386)	(4.787)	(8.862)
R-squared	0.9968	0.9968	0.997	0.9968
Obs.	126480	111024	95616	80256

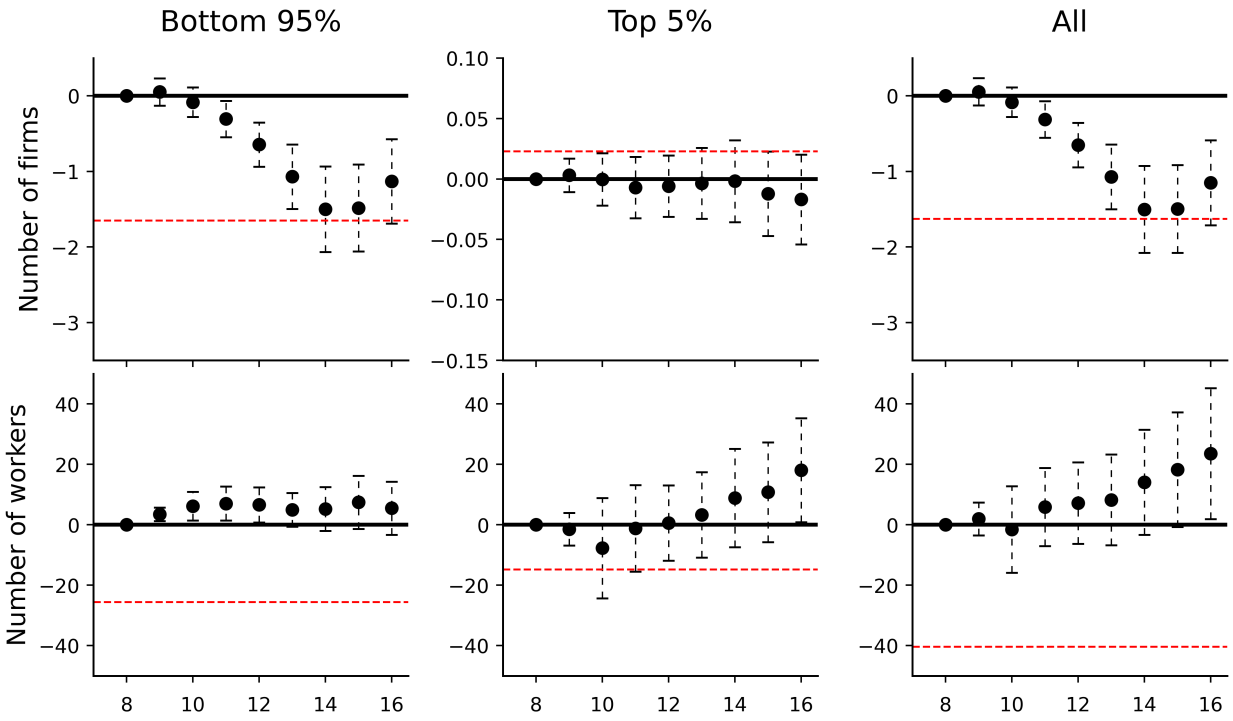
Notes: This table reports the triple-differences estimates examining the effect of the truce and several placebo periods set 3, 5, 7, and 9 months prior to the actual onset of the truce. Each column presents results from a separate specification. The "Treatment" column presents the baseline top-half homicide-reduction split, while the lower, middle, and upper thresholds apply the same cutoffs used in the threshold-specific exercises, defined by the 25th, 50th, and 75th percentiles of the homicide-reduction distribution among treated polygons. The rows labeled "Treatment (Placebo)" report the pooled difference-in-differences estimates for assignment dates preceding the actual truce. The row "Treatment x Truce" reports the incremental estimate for the actual-truce scenario relative to those placebo scenarios. Standard errors (in parentheses) are clustered at the polygon level interacted with placebo period or truce. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Figure B.9: Sectoral Growth and Economic Proportions During the Truce



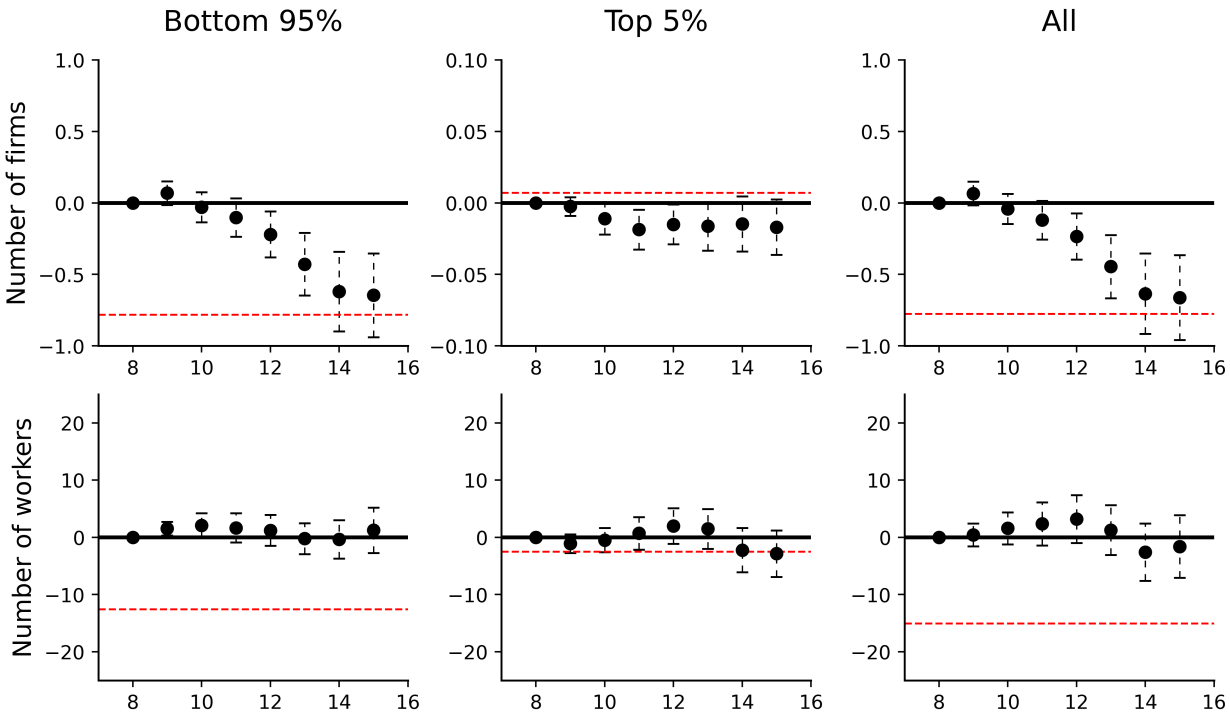
Notes: This graph presents a three-part analysis based on different sectors as classified by the CIIU Rev. 3 standard. The first column represents baseline regression estimates, calculated using sector-specific outcome variables to assess the impact of the truce on each sector. The second column indicates the relative growth in each sector due to the truce, calculated as the treatment effect estimator divided by the pre-truce levels of the corresponding indicator. The third column provides context by showing the percentage that each sector contributes to the overall economy just prior to the truce. The sample excludes the top 5% of firms.

Figure B.10: Post-Truce Event-Study Estimates by Firm Size: Measured Gang Exposure



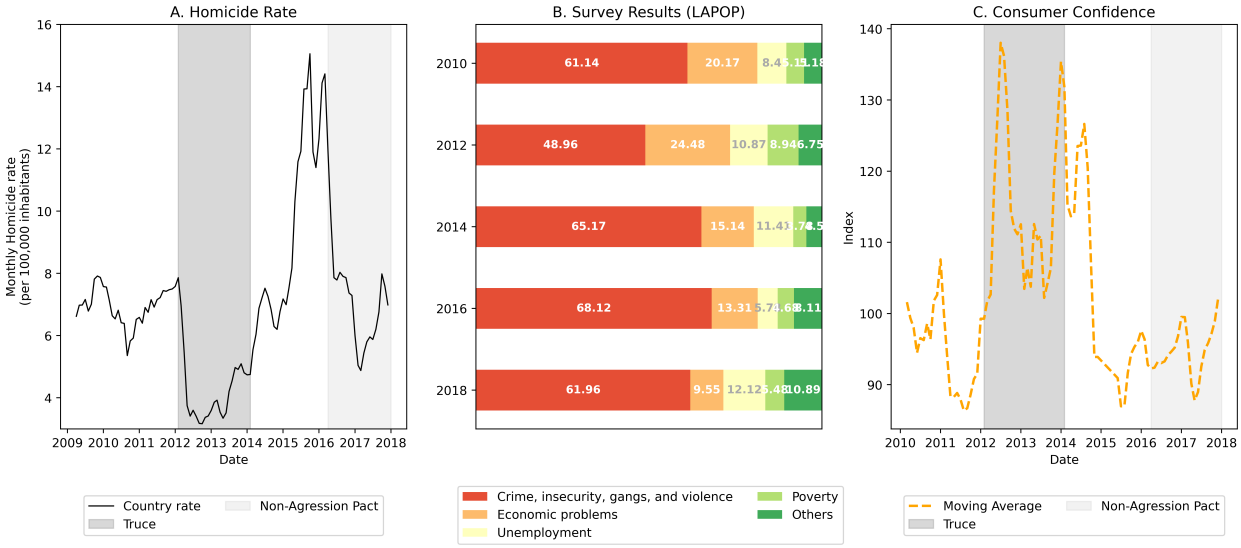
Notes: This figure presents event-study estimates following the end of the gang truce, using a specification analogous to Equation (1)*. Exposure is defined using measured pre-truce gang-related homicide activity. Outcomes are shown separately for small and medium firms (left panel), large firms (center panel), and all firms combined (right panel). Time is aggregated into three-month intervals after the end of the truce. The red dotted horizontal line represents the estimated outcome difference in the pre-truce reference period. All regressions include unit and time fixed effects. Vertical bars represent 95% confidence intervals, with standard errors clustered at the geographic unit level.*

Figure B.11: Post-Truce Event-Study Estimates by Firm Size: Homicide-Reduction Classification



Notes: This figure presents event-study estimates following the end of the gang truce, using the complementary homicide-reduction classification from the previous specification. The classification compares polygons in the top 50% of realized homicide reductions during the truce to polygons in the bottom 50%. Outcomes are shown separately for small and medium firms (left panel), large firms (center panel), and all firms combined (right panel). Time is aggregated into three-month intervals spanning 24 months after the truce. The red dotted horizontal line represents the average level of the outcome variable immediately prior to the truce. All regressions include unit and time fixed effects. Vertical bars represent 95% confidence intervals, with standard errors clustered at the geographic unit level.

Figure B.12: Crime, Public Perceptions, and Consumer Confidence During the Truce and Non-Aggression Pact Periods



Notes: This figure shows descriptive evidence related to the 2012 gang truce and subsequent non-aggression pact periods in El Salvador. ***Panel A*** *displays the national monthly homicide rate (per 100,000 inhabitants), aggregated from administrative data and smoothed with a 3-month moving average.* ***Panel B*** *illustrates the distribution of survey responses to a question on the most pressing problems facing the country, using LAPOP nationally representative household data.* ***Panel C*** *shows the evolution of the Consumer Confidence Index. The shaded gray areas indicate two key episodes: the gang truce (March 2012) and the non-aggression pact (April 2016).*

B.4 Theoretical Framework

This appendix develops a parsimonious model that organizes the mechanisms suggested by the data. The economy is static, competitive, and features two geographic zones $z \in \{H, L\}$ producing a local non–tradable. For clarity in the exposition, whenever referring to the *low-crime zone* in the text I write *zone* L; symbols with subscript L (e.g., q_L, P_L) remain as standard notation.

The setup features two deviations from a standard local–demand framework: (i) crime enters the *consumer side* as a fear-cost wedge that raises the effective price of purchasing locally in the high-crime zone H; and (ii) crime also shifts the *labor-supply side* by lowering participation at a given wage (crime-sensitive participation margin).

B.4.1 Environment and Agents

Time is static. There is a representative household of unit mass and a competitive sector of firms in each zone. Labor is perfectly mobile across zones and the production technology is linear in labor with common productivity $A > 0$.

Zones and crime. I index zones by $z \in \{H, L\}$ and let $c \geq 0$ denote the (exogenous) crime level specific to the high-crime zone H. For parsimony I normalize crime in the low-crime zone L to zero; allowing $c_{\mathsf{L}} > 0$ is a straightforward variation.

B.4.2 Firms and Technology

Firms in each zone hire local labor ℓ_z and produce local output y_z using

$$y_z \;=\; A\,\ell_z, \qquad A > 0.$$

Perfect competition and free mobility of labor imply that the (common) producer price equals unit cost,

$$p \;=\; \frac{w}{A},$$

where w is the common market wage. Local goods are non–tradable, so market clearing in each zone requires $y_z = q_z$, equivalently $A\,\ell_z = q_z$.

B.4.3 Goods, Crime, and Effective Prices (numéraire iceberg; household closure)

Crime affects consumption in H via a reduced-form fear-cost wedge $\chi(c)$ that raises the *effective consumer price* paid by households:

$$P_L \equiv p, \qquad P_H \equiv p + \chi(c), \qquad \chi(c) = \beta\, c, \;\; \beta > 0.$$

Household income, utility, and budget. Preferences are quasilinear in the numéraire:

$$U \;=\; a_H \ln q_H \;+\; a_L \ln q_L \;+\; m, \qquad a_H, a_L > 0.$$

The parameters a_H and a_L govern demand for goods in each zone, and m denotes consumption of the numéraire good. Total numéraire income is

$$Y \;\equiv\; \bar{Y} \;+\; w\, L, \qquad \bar{Y} \geq 0,$$

where $\bar{Y}$ is an exogenous endowment/transfer and wL is labor income (firms earn zero profits under perfect competition and $y_z = A\ell_z$). The representative household chooses (q_H, q_L, m) subject to

$$m \;+\; P_H\, q_H \;+\; P_L\, q_L \;=\; Y.$$

Resource accounting (numéraire sink). Of the expenditure $P_H q_H$, firms receive $p\, q_H$; the residual $\chi(c)\, q_H$ is numéraire *used and destroyed* (no agent receives it). Aggregating,

$$\bar{Y} \;+\; wL \;=\; m \;+\; p\,(q_H + q_L) \;+\; \chi(c)\, q_H,$$

and with $p = \frac{w}{A}$ and local clearing $q_z = A\ell_z$, we have $p(q_H + q_L) = wL$, so the accounting identity holds.

Feasibility (numéraire nonnegativity). Using $q_H^* = \dfrac{a_H}{p^* + \chi(c)} = \dfrac{a_H A}{w^* + A\beta c}$,

$$m^* \;=\; \bar{Y} \;-\; \chi(c)\, q_H^* \;=\; \bar{Y} \;-\; \beta c \cdot \frac{a_H A}{w^* + A\beta c}.$$

Thus the interior allocation is feasible provided

$$\boxed{\bar{Y} \;\geq\; \chi(c)\, q_H^* \;=\; \beta c \cdot \frac{a_H A}{w^* + A\beta c}}.$$

In what follows we assume $\bar{Y}$ is large enough so that $m^* \geq 0$. Because preferences are quasilinear, q_H and q_L depend only on (p, χ); m adjusts residually.

Demand. Utility maximization yields the Marshallian demands used throughout:

$$q_H = \frac{a_H}{p + \chi(c)}, \qquad q_L = \frac{a_L}{p}.$$

Remark on "closed economy." "Closed" here refers to the local non-tradable good and internal wage/price determination: goods and labor markets clear within the economy. The numéraire is modeled as an endowment/transfer $\bar{Y}$ financing residual consumption m.

B.4.4 Labor Supply and Demand

Labor Supply. Crime also affects participation. I model this with a linear shift in the labor-supply schedule:

$$L^S(w, c) \;=\; s\,[\,w - \psi(c)\,]_+, \qquad \psi(c) \equiv \gamma c, \quad s > 0,\ \gamma > 0,$$

where $[x]_+ = \max\{x, 0\}$. Parameter s is the slope of labor supply with respect to the wage, and γ determines how crime raises the participation threshold through commuting risk, time losses, and safety concerns.

Labor demand. Using $A\ell_z = q_z$, $p = w/A$, and $\chi(c) = \beta c$, the zone-level and aggregate labor demands are

$$\ell_H(w, c) = \frac{a_H}{w + A\beta c}, \qquad \ell_{\mathsf{L}}(w) = \frac{a_L}{w}, \qquad L^D(w, c; a_H, a_L) = \frac{a_H}{w + A\beta c} + \frac{a_L}{w}.$$

Interior branch. For any $w \in (0, \gamma c]$, $L^S(w, c) = 0$ while $L^D(w, c; a_H, a_L) > 0$. At the participation boundary, $L^D(\gamma c, c) \in (0, \infty)$ if $c > 0$; and for any $c \geq 0$, $\lim_{w \downarrow 0} L^D(w, c) = +\infty$. Hence the market-clearing condition $L^S = L^D$ cannot hold at or below the participation threshold, and any equilibrium must satisfy $w^* > \gamma c$. On the interior domain $(\gamma c, \infty)$, $L^S(w, c) = s(w - \gamma c)$ is differentiable and strictly increasing, which I use in the existence and uniqueness proof below.

B.5 Competitive equilibrium: definition, existence, and uniqueness

I now characterize the competitive equilibrium for the baseline specification with size–only local demand and crime–sensitive labor supply introduced above.

B.5.1 Definition of equilibrium

Given parameters $(A, a_H, a_L, s, \beta, \gamma)$ and the crime level $c \geq 0$, a (competitive) equilibrium is a wage $w^* > 0$ and associated allocations

$$p^* = \frac{w^*}{A}, \qquad P_H^* = p^* + \beta c, \qquad P_L^* = p^*,$$

$$q_H^* = \frac{a_H}{P_H^*} = \frac{a_H A}{w^* + A\beta c}, \qquad q_L^* = \frac{a_L}{P_L^*} = \frac{a_L A}{w^*},$$

$$\ell_H^* = \frac{q_H^*}{A} = \frac{a_H}{w^* + A\beta c}, \qquad \ell_L^* = \frac{q_L^*}{A} = \frac{a_L}{w^*},$$

such that local non–tradable markets clear ($y_z^* = q_z^*$) and aggregate labor supply equals aggregate labor demand:

$$L^S(w^*, c) \;=\; L^D(w^*, c; a_H, a_L),$$

$$L^S(w, c) = s\,[\,w - \gamma c\,]_+, \qquad L^D(w, c; a_H, a_L) = \frac{a_H}{w + A\beta c} + \frac{a_L}{w}. \tag{E}$$

I focus on the empirically relevant interior case $w^* > \gamma c$, under which $L^S(w, c) = s(w - \gamma c)$ is differentiable.

B.5.2 Existence and uniqueness (baseline size–only)

Lemma B.5.1 (Properties and boundary behavior of L^D) *For any $a_H, a_L, A > 0$, $\beta > 0$, and finite $c \geq 0$, the aggregate labor demand*

$$L^D(w, c; a_H, a_L) = \frac{a_H}{w + A\beta c} + \frac{a_L}{w}$$

is continuous on $(0, \infty)$, strictly decreasing in w, and satisfies

$$\lim_{w \downarrow 0} L^D(w, c) = +\infty, \qquad \lim_{w \uparrow \infty} L^D(w, c) = 0, \qquad \frac{\partial L^D}{\partial w}(w, c) = -\frac{a_H}{(w + A\beta c)^2} - \frac{a_L}{w^2} < 0.$$

Moreover, at the participation boundary $w = \gamma c$:

$$\begin{cases} L^D(\gamma c, c) = \dfrac{a_H}{(\gamma + A\beta)c} + \dfrac{a_L}{\gamma c} \in (0, \infty), & \text{if } c > 0, \\ \lim\limits_{w \downarrow 0} L^D(w, 0) = +\infty, & \text{if } c = 0. \end{cases}$$

Proof. All statements follow directly from the closed form and elementary limits. ■

Lemma B.5.2 (Properties of L^S on the interior branch) *For $w > \gamma c$, $L^S(w, c) = s(w - \gamma c)$ is continuous and strictly increasing in w, with $L^S(\gamma c, c) = 0$ and $\lim_{w \uparrow \infty} L^S(w, c) = +\infty$.*

Proof. Immediate from linearity with $s > 0$. ■

Proposition 1 (Existence and uniqueness) *Fix $a_H, a_L, A, s > 0$, $\beta, \gamma > 0$, and finite $c \geq 0$. There exists a unique $w^* > \gamma c$ solving* (E).

Proof. Let $H(w; c) \equiv L^D(w, c; a_H, a_L) - L^S(w, c)$ for $w \geq \gamma c$.

Case $c > 0$. At $w = \gamma c$,

$$L^S(\gamma c, c) = 0, \qquad L^D(\gamma c, c) = \frac{a_H}{(\gamma + A\beta)c} + \frac{a_L}{\gamma c} > 0,$$

so $H(\gamma c; c) > 0$.

Case $c = 0$. For $w > 0$, $H(w; 0)$ is continuous and

$$\lim_{w \downarrow 0} H(w; 0) = \lim_{w \downarrow 0} \Big(\frac{a_H}{w} + \frac{a_L}{w} - s\, w \Big) = +\infty,$$

since $L^D(w, 0) = (a_H + a_L)/w \to +\infty$ and $L^S(w, 0) = s\, w \to 0$.

In both cases H is continuous on $(\gamma c, \infty)$ and

$$\lim_{w \to \infty} H(w; c) = \lim_{w \to \infty} \big(L^D(w, c) - L^S(w, c) \big) = 0 - (+\infty) = -\infty.$$

By the Intermediate Value Theorem, there exists $w^* > \gamma c$ with $H(w^*; c) = 0$ (existence). Uniqueness follows because $L^D(\cdot, c)$ is strictly decreasing and $L^S(\cdot, c)$ is strictly increasing on $(\gamma c, \infty)$, so the two schedules can cross at most once. ■

Interiorness and feasibility. Proposition 1 implies $w^* > \gamma c$, i.e., the interior branch is the relevant equilibrium (the kink is not binding). Given quasilinear preferences, the Marshallian quantities are

income–independent; I assume the numéraire endowment $\bar{Y}$ is sufficiently large so that the implied $m^* \geq 0$ (feasibility of the interior allocation).

Equilibrium objects (summary). The unique equilibrium is characterized by the scalar w^* that solves (E). All remaining prices and allocations follow from

$$p^* = \frac{w^*}{A}, \quad P_H^* = p^* + \beta c, \quad P_L^* = p^*, \qquad (q_H^*, q_L^*) = \Big(\frac{a_H A}{w^* + A\beta c}, \frac{a_L A}{w^*}\Big), \qquad (\ell_H^*, \ell_L^*) = \Big(\frac{a_H}{w^* + A\beta c}, \frac{a_L}{w^*}\Big).$$

These expressions will be used below to derive the comparative statics with respect to crime.

B.6 Crime and equilibrium outcomes: wage and aggregate employment

We now study how the competitive equilibrium reacts to changes in crime c in the high–crime zone H, focusing on the equilibrium wage $w^*(c)$ and aggregate employment $L^*(c)$. Throughout we retain the size–only demand and crime–sensitive labor supply of (E), and the interior branch $w^* > \gamma c$.

B.6.1 Setup for comparative statics

Let

$$F(w, c) \;\equiv\; s\,[\,w - \gamma c\,] \;-\; \frac{a_H}{w + A\beta c} \;-\; \frac{a_L}{w},$$

so that the equilibrium condition is $F(w^*(c), c) = 0$. Define the convenient shorthands

$$X \;\equiv\; w^*(c) + A\beta c, \qquad Z \;\equiv\; w^*(c).$$

Direct differentiation yields

$$F_w \;=\; s + \frac{a_H}{X^2} + \frac{a_L}{Z^2} \;>\; 0, \qquad F_c \;=\; -\,s\gamma \;+\; \frac{a_H A\beta}{X^2}.$$

By the Implicit Function Theorem,

$$\frac{dw^*}{dc} \;=\; -\,\frac{F_c}{F_w} \;=\; -\,\frac{-s\gamma + \dfrac{a_H A\beta}{X^2}}{s + \dfrac{a_H}{X^2} + \dfrac{a_L}{Z^2}}. \tag{10}$$

B.6.2 Equilibrium wage

Proposition 2 (Wage response to crime) *Under the baseline specification, the equilibrium wage is (locally) decreasing in crime if and only if*

$$\frac{a_H A\beta}{X^2} \;>\; s\,\gamma. \tag{C1}$$

Equivalently, when (C1) *holds, a reduction in crime ($\Delta c < 0$) raises the wage ($\Delta w^* > 0$).*

Interpretation. Condition (C1) compares the *demand-side* impact of crime (the fear-cost wedge in H, scaled by its market size a_H and productivity A) to the *supply-side* inward shift (crime-sensitive participation $s\gamma$). If the demand effect dominates, crime reductions push the wage up; otherwise the wage response can be attenuated or (locally) flat/negative.

B.6.3 Aggregate employment

Aggregate employment equals

$$L^*(c) \;=\; L^S(w^*(c), c) \;=\; s\,[\,w^*(c) - \gamma c\,].$$

Differentiating and using (10) gives

$$\begin{aligned}\frac{dL^*}{dc} &= s\Big(\frac{dw^*}{dc} - \gamma\Big) = s\,\frac{-\big(-s\gamma + \frac{a_H A\beta}{X^2}\big) \;-\; \gamma\big(s + \frac{a_H}{X^2} + \frac{a_L}{Z^2}\big)}{s + \frac{a_H}{X^2} + \frac{a_L}{Z^2}} \\ &= -\,s\,\frac{a_H\,\dfrac{A\beta + \gamma}{X^2} \;+\; a_L\,\dfrac{\gamma}{Z^2}}{s + \dfrac{a_H}{X^2} + \dfrac{a_L}{Z^2}}.\end{aligned}$$

Proposition 3 (Crime strictly lowers aggregate employment) *Fix $a_H, a_L, s, A > 0$ with $\beta > 0$ and $\gamma > 0$. Then*

$$\frac{dL^*}{dc} \;<\; 0.$$

Equivalently, any reduction in crime ($\Delta c < 0$) strictly *increases aggregate employment ($\Delta L^* > 0$).*

Intuition. A crime reduction shifts *demand* outward by lowering the effective price in H and shifts *supply* outward by raising participation. Even if the wage response is muted (e.g., supply expands strongly), the outward movements ensure that total employment rises unambiguously.

B.7 Zone-by-zone effects: substitution across zones vs. participation gains

We now decompose the employment response by zone when crime c changes in H. Recall (size–only demand, crime–sensitive supply):

$$\ell_H = \frac{a_H}{w + A\beta c}, \qquad \ell_{\mathsf{L}} = \frac{a_L}{w}, \qquad L^* = s\,[\,w^*(c) - \gamma c\,],$$

and define the shorthands

$$X \equiv w^* + A\beta c, \qquad Z \equiv w^*.$$

Local derivatives and a convenient decomposition. Total derivatives with respect to c can be written as

$$\frac{d\ell_H}{dc} = \underbrace{\Big(-\frac{a_H A\beta}{X^2}\Big)}_{\text{direct demand (crime wedge in } H\text{) at fixed } w} + \underbrace{\Big(-\frac{a_H}{X^2}\Big)\frac{dw^*}{dc}}_{\text{wage (general-equilibrium) channel}}, \tag{11}$$

$$\frac{d\ell_{\mathsf{L}}}{dc} = \underbrace{0}_{\text{no direct crime term in the low-crime zone}} + \underbrace{\Big(-\frac{a_L}{Z^2}\Big)\frac{dw^*}{dc}}_{\text{wage (general-equilibrium) channel}}. \tag{12}$$

Thus, any change in employment in the low-crime zone is *purely* a reallocation (via the wage), while the high-crime zone combines a direct local-demand force with the wage channel.

Using the equilibrium condition (Section B.5), the wage derivative is

$$\frac{dw^*}{dc} = -\,\frac{-s\gamma + \dfrac{a_H A\beta}{X^2}}{s + \dfrac{a_H}{X^2} + \dfrac{a_L}{Z^2}} \equiv -\,\frac{F_c}{F_w}, \qquad F_w > 0. \tag{13}$$

B.7.1 When does cross-zone substitution occur?

A *crime reduction* ($\Delta c < 0$) raises aggregate employment L^* (Proposition 3). Whether we also observe reallocation away from the low-crime zone L hinges on the wage response.

Low-crime zone monotonicity. Employment in the low-crime zone is $\ell_{\mathsf{L}}(w) = a_L/w$, which is strictly decreasing in w. Thus, for any finite change from c to c' with $w_1 = w^*(c) > 0$ and $w_2 = w^*(c') > 0$,

$$\Delta\ell_{\mathsf{L}} = \ell_{\mathsf{L}}(w_2) - \ell_{\mathsf{L}}(w_1) = a_L\Big(\frac{1}{w_2} - \frac{1}{w_1}\Big) = -\,a_L\,\frac{\Delta w^*}{w_1 w_2}, \qquad \Delta w^* \equiv w_2 - w_1,$$

so

$$\operatorname{sign}(\Delta \ell_{\mathsf{L}}) = -\operatorname{sign}(\Delta w^*).$$

Local vs. finite changes. At the margin, reallocation from L occurs iff the wage increases when crime falls:

$$dc < 0 \;\Rightarrow\; dw^* > 0 \iff d\ell_{\mathsf{L}} < 0.$$

In our baseline, this local wage increase holds *iff* condition

$$\frac{a_H A\beta}{X^2} \;>\; s\,\gamma \quad\Longleftrightarrow\quad \frac{dw^*}{dc} < 0, \tag{C1}$$

with $X \equiv w^* + A\beta c$. For a *finite* crime change Δc, the global implication $\Delta c < 0 \Rightarrow \Delta w^* > 0 \Rightarrow \Delta \ell_{\mathsf{L}} < 0$ follows provided the sign $dw^*/dc < 0$ is maintained along the entire path in c (equivalently, **(C1)** holds for all intermediate crime levels).

Why ℓ_H must rise. Adding up $L^* = \ell_H + \ell_{\mathsf{L}}$ and using $\Delta L^* > 0$:

- If **(C1)** holds, then $\Delta \ell_{\mathsf{L}} < 0$ and thus

$$\Delta \ell_H = \Delta L^* - \Delta \ell_{\mathsf{L}} > 0.$$

- If **(C1)** fails, the wage is weakly non-increasing, so the direct reduction in the H-wedge (lower $A\beta c$ in $w^* + A\beta c$) together with a flat/declining wage implies $\Delta \ell_H > 0$.

Hence, with a crime reduction, employment in H rises unambiguously; whether employment in L falls depends solely on the sign of the wage response via **(C1)**.

Formal proof. Let $X \equiv w^* + A\beta c$ and $Z \equiv w^*$. Zone-H employment is

$$\ell_H \;=\; \frac{a_H}{X} \quad\Longrightarrow\quad \frac{d\ell_H}{dc} \;=\; -\,\frac{a_H}{X^2}\Big(\frac{dw^*}{dc} + A\beta\Big).$$

Using the equilibrium condition $F(w, c) = 0$ with

$$F_w \;=\; s + \frac{a_H}{X^2} + \frac{a_L}{Z^2} \;>\; 0, \qquad F_c \;=\; -\,s\gamma + \frac{a_H A\beta}{X^2},$$

the Implicit Function Theorem gives

$$\frac{dw^*}{dc} = -\frac{F_c}{F_w} \quad \Rightarrow \quad \frac{dw^*}{dc} + A\beta = \frac{-F_c + A\beta\, F_w}{F_w} = \frac{s(\gamma + A\beta) + A\beta\, \frac{a_L}{Z^2}}{F_w} \;>\; 0.$$

Therefore $\frac{d\ell_H}{dc} < 0$, i.e., a reduction in crime ($\Delta c < 0$) *strictly* increases employment in H:

$$\Delta c < 0 \;\Rightarrow\; \Delta \ell_H > 0.$$

B.7.2 An intuitive reallocation index

To keep the interpretation simple, summarize the composition of the employment gain from a crime reduction ($\Delta c < 0$) with a single *intensity index*:

$$\Delta L^* \;=\; \Delta \ell_H \;+\; \Delta \ell_{\mathsf{L}}, \qquad \text{with } \; \Delta L^* > 0 \text{ under } \beta, \gamma > 0.$$

Definition (reallocation intensity).

$$\boxed{\text{Reallocation index } \; I_{\text{sub}} \;\equiv\; \frac{\max\{0,\, -\Delta \ell_{\mathsf{L}}\}}{\Delta L^*}.}$$

When is reallocation active? Reallocation is active ($I_{\text{sub}} > 0$) if and only if the wage rises with the crime reduction, which occurs under the dominance condition

$$\frac{a_H A\beta}{X^2} \;>\; s\,\gamma \quad \Longleftrightarrow \quad \Delta w^* > 0 \quad \Longleftrightarrow \quad \Delta \ell_{\mathsf{L}} < 0. \tag{C1}$$

How to read the index.

- **Large I_{sub} (reallocation–intensive):** many jobs leave L relative to the net gain. Typical when a_H is large, the fear–cost channel β is strong (or $X = w^* + A\beta c$ is low), and supply sensitivity $s\gamma$ is modest. Empirically: wage up, ℓ_{L} down.
- **Small I_{sub} (participation–driven):** few jobs leave L compared to the net gain. Typical when s and/or γ are high (wage moves little or falls), or when the base in L is small (a_L small / $Z = w^*$ large). Empirically: wage flat/down, ℓ_{L} flat/up.